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Automotive Acoustic Simulation and Siren Signal Localization

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Hamburg, Juni 2025

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Declaration of Academic Honesty

I, AMR AHMED, student of the Master of Science in Mechatronics at Technische Universität Hamburg (TUHH), matriculation number 568447, hereby declare that I have written the present Master's thesis independently and without unauthorized assistance. I have used only the sources and aids explicitly cited in the text and listed in the bibliography.

This thesis has not been submitted, either in the same or a similar form, to any other examination authority.

Hamburg, June 17, 2025

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Nomenclature

List of Symbols

Latin List of Symbols

Symbol	Unit	Meaning
$a(i)$	-	Steering vector for grid point i
$a_{\text{geometric}}$	-	Geometric attenuation factor
$a_{\text{atmospheric}}$	-	Atmospheric attenuation factor
a_{total}	-	Total attenuation factor
a_k	-	Feedback coefficients in IIR filter
A	-	Array aperture matrix / Denominator polynomial
b_{CBF}	Pa ²	Conventional beamforming output
b_k	-	Feedforward coefficients in IIR filter
B	Pa ²	Beamforming output matrix
$B(z)$	-	Numerator polynomial in digital filter transfer function
$A(z)$	-	Denominator polynomial in digital filter transfer function
c	m/s	Speed of sound in medium (typically 343 m/s in air)
$c_i^{(k)}$	-	Correlation values in OMP-DAMAS algorithm
CSM	Pa ²	Cross-Spectral Matrix
D	-	Doppler factor
D_{smooth}	-	Smoothed Doppler factor
d	m	Inter-element spacing / Distance
$d_j(t_s)$	m	Distance for reflection path j at source time t_s
e	-	Euler's number (2.718...)
F	-	Array figure of merit (optimization parameter) / Support set
f	Hz	Frequency
f_c	Hz	Cutoff frequency
f_s	Hz	Sampling frequency
$f_{s,o}$	Hz	Oversampling frequency
$f_{s,r}$	Hz	Receiver sampling frequency
f_b	Hz	Signal bandwidth
f_N	Hz	Nyquist frequency
f_{over}	Hz	Oversampling frequency
G	1/m	Green's function
$G_{xy}(\omega)$	-	Cross-power spectral density

Latin List of Symbols - Continue

Symbol	Unit	Meaning
$G_{xx}(\omega)$	-	Auto-power spectral density of signal x
$G_{yy}(\omega)$	-	Auto-power spectral density of signal y
H	-	Frequency-domain transfer function
$H(z)$	-	Digital transfer function
$H_a(j\omega)$	-	Analog transfer function
$h(t)$	-	Room impulse response function
h_{final}	-	Final processed impulse response
h_{raw}	-	Raw extracted impulse response
I	-	Identity matrix
$i_{\text{max}}^{(k)}$	-	Index of maximum correlation in iteration k
j	-	Imaginary unit ($\sqrt{-1}$)
k	1/m	Wavenumber ($k = \omega/c$)
K	-	Sparsity level / Maximum iterations
K_{max}	-	Maximum number of iterations
L	-	Step size in SC-DAMAS algorithm / Signal length
L_x	-	Length of signal x
L_y	-	Length of signal y
L_{filtered}	-	Length of filtered signal
M	-	Number of microphones
$n(t)$	-	Noise signal
$n_m(t)$	-	Noise signal at microphone m
N	-	Oversampling ratio / Signal length
N_{grid}	-	Number of grid points
N_{min}	-	Minimum oversampling ratio
N_{base}	-	Base oversampling factor
$N_{\text{oversample}}$	-	Oversampling factor
N_{chunk}	-	Chunk size for processing
N_{paths}	-	Number of reflection paths
n_{max}	-	Maximum reflection order
n_x, n_y, n_z	-	Reflection indices along spatial axes
p	Pa	Acoustic pressure
p_s	Pa	Source signal pressure
p_{over}	Pa	Oversampled pressure signal
p_{filtered}	Pa	Filtered pressure signal
p_{direct}	Pa	Direct path pressure signal
$p_{\text{noisy},m}$	Pa	Noisy signal at microphone m

Latin List of Symbols - Continue

Symbol	Unit	Meaning
$p_{\text{clean},m}$	Pa	Clean signal at microphone m
P_{CBF}	Pa ²	Conventional beamforming power
P_{true}	Pa ²	True source power distribution
$P_{\text{signal},m}$	Pa ²	Signal power at microphone m
PSF	-	Point Spread Function matrix
q	-	Index for reflection computation
Q	-	Safety factor for filtering
r	m	Distance
r_m	m	Distance/position vector of microphone m
r_s	m	Source position
r_r	m	Receiver position
$\mathbf{r}_s(t_e)$	m	Source position at emission time
$\mathbf{r}_r$	m	Receiver position vector
$\mathbf{r}_m$	m	Microphone position vector
$\mathbf{R}_p$	m	Distance from source to reflection point
$\mathbf{R}_r$	m	Distance from reflection point to receiver
R_{norm}	-	Normalized cross-correlation
RT60	s	Reverberation time
s	-	Laplace domain variable
$\mathbf{s}(t)$	m	Source trajectory
$\mathbf{s}_j(t)$	m	Mirror source trajectory for path j
$S_{\text{ref}}(t)$	-	Reference exponential sine sweep
SF	-	Spectral flatness
SNR _{dB}	dB	Signal-to-noise ratio in decibels
t	s	Time
t_e	s	Emission time
t_r	s	Reception time
t_s	s	Source time
$t_{r,j}(t_s)$	s	Arrival time for reflection path j
t_{over}	s	Oversampled time grid
t_{min}	s	Minimum time
t_{max}	s	Maximum time
t_{start}	s	Template start time
$t_{\text{chunk},m}$	s	Time chunk m
T	s	Template signal / Sampling period
T_{duration}	s	Template duration

Latin List of Symbols - Continue

Symbol	Unit	Meaning
u	-	Correlation vector
$\hat{u}$	-	Unit vector
v	m/s	Velocity
$\vec{v}$	m/s	Velocity vector
v_{rel}	m/s	Relative radial velocity
W	-	Window half-width / Window size
w_{causal}	-	Causality window function
w_{fade}	-	Fade-out window function
w_{denoise}	-	Denoising window function
$x(t)$	-	Source signal
$x_m(t)$	-	Signal at microphone m
x_{recorded}	-	Recorded signal
$x_{\text{simulated}}$	-	Simulated signal
$\mathbf{x}_F$	-	Solution vector restricted to support set F
$\mathbf{x}_{\text{OMP}}$	-	OMP solution vector
$X(\omega)$	-	Fourier transform of source signal
$y(t)$	-	Received acoustic signal
$y_\theta(t)$	-	Beamformed output for angle θ
$Y(\omega)$	-	Fourier transform of received signal
$Y_j(f)$	-	Frequency response for reflection path j
$Y_{\text{total}}(f)$	-	Total frequency response
z	-	Z-domain variable

Greek List of Symbols

Symbol	Unit	Meaning
α	-	Decay factor in learning rate schedule / Atmospheric absorption coefficient / Signal-dependent scaling factor
β	-	Reflection coefficient
β_{SF}	-	Spectral flatness correction factor
β_{total}	-	Combined reflection coefficient
$\beta_{\text{total},j}$	-	Combined reflection coefficient for path j
β_{x1}, β_{x2}	-	Reflection coefficients for x-direction walls
β_{y1}, β_{y2}	-	Reflection coefficients for y-direction walls
β_{z1}, β_{z2}	-	Reflection coefficients for z-direction walls
γ	-	Absorption coefficient
γ_{x1}, γ_{x2}	-	Absorption coefficients for x-direction walls
γ_{y1}, γ_{y2}	-	Absorption coefficients for y-direction walls
γ_{z1}, γ_{z2}	-	Absorption coefficients for z-direction walls
δ	-	Dirac delta function
δ_K	-	Small constant in SC-DAMAS algorithm
ϵ	-	Regularization parameter / Small constant
ϵ_{res}	-	Residual threshold
ϵ_{corr}	-	Correlation threshold
θ	rad	Azimuth angle / Steering angle
θ_{source}	rad	Estimated source angle
λ	-	Regularization parameter / Wavelength
$\lambda(\omega)$	-	Frequency-dependent regularization parameter
λ_{reg}	-	Tikhonov regularization parameter
μ	-	Mean value
ν	-	Frequency (alternative notation)
ξ	-	Damping ratio
ρ	-	Correlation coefficient / Density
$\rho_{\text{verification}}$	-	Verification correlation coefficient
σ	-	Standard deviation
σ^2	-	Variance
σ_{signal}^2	-	Signal variance
$\sigma_{\text{quantization}}^2$	-	Quantization error variance
σ_{recorded}	-	Standard deviation of recorded signal
$\sigma_{\text{simulated}}$	-	Standard deviation of simulated signal
τ	s	Time delay / Threshold in sparse recovery / Temporal lag
τ_{optimal}	s	Optimal time alignment
τ_m	s	Time delay for microphone m
τ_j	s	Propagation delay for path j
ϕ	rad	Elevation angle

Greek List of Symbols

Symbol	Unit	Meaning
ψ	-	Phase
ω	rad/s	Angular frequency ($\omega = 2\pi f$)
ω_c	rad/s	Cutoff angular frequency
$\omega_{c,\text{norm}}$	-	Normalized cutoff frequency
Ω	-	Support set in sparse recovery
$\Omega^{(k)}$	-	Support set at iteration k

Indexes

Index	=	Meaning
e	=	Emission time
r	=	Receiver/reception time
s	=	Source
i, j	=	Grid point indices / General indices
k	=	Iteration index / Frequency bin index / Sample index
l	=	Sample index (alternative)
m	=	Microphone index
n	=	Sample index
n_x, n_y, n_z	=	Integer set related to reflection order
p	=	Reflection path index
q	=	Index for reflection computation
over	=	Oversampled
raw	=	Unprocessed value
final	=	Processed/smoothed value
cbf	=	Conventional beamforming
smooth	=	Smoothed value
max	=	Maximum value
min	=	Minimum value
opt	=	Optimal value
ref	=	Reference value
start	=	Starting value
end	=	Ending value
total	=	Combined/total value
L, R	=	Left and right ears/microphones
OMP	=	Orthogonal Matching Pursuit
SC	=	Sparsity Constrained
SF	=	Spectral Flatness
res	=	Residual
corr	=	Correlation
reg	=	Regularization
dB	=	Decibels
norm	=	Normalized

Abbreviations

Abbr.	=	Meaning
ANE	=	Anomalous Noise Events
API	=	Application Programming Interface
CBF	=	Conventional Beamforming
CFD	=	Computational Fluid Dynamics
CNN	=	Convolutional Neural Network
CoSaMP	=	Compressive Sampling Matching Pursuit
CSM	=	Cross-Spectral Matrix
DAMAS	=	Deconvolution Approach for the Mapping of Acoustic Sources
DNN	=	Deep Neural Network
DOA	=	Direction of Arrival
DRR	=	Direct-to-Reverberant Ratio
EDT	=	Early Decay Time
FFT	=	Fast Fourier Transform
FIR	=	Finite Impulse Response
GPU	=	Graphics Processing Unit
HD	=	High Definition
IEC	=	International Electrotechnical Commission
IIR	=	Infinite Impulse Response
ILD	=	Interaural Level Difference
ITD	=	Interaural Time Difference
MAE	=	Mean Absolute Error
MEMS	=	Microelectromechanical Systems
MSL	=	Maximum Sidelobe Level
NOTAP	=	N-Oversampled Transmitted Acoustic Pressure
OMP	=	Orthogonal Matching Pursuit
PCHIP	=	Piecewise Cubic Hermite Interpolation
PINN	=	Physics-Informed Neural Network
PSF	=	Point Spread Function
ReLU	=	Rectified Linear Unit
REPL	=	Read-Eval-Print Loop
RIR	=	Room Impulse Response
RMS	=	Root Mean Square
RMSE	=	Root Mean Square Error
SAMP	=	Sparsity Adaptive Matching Pursuit
SC-DAMAS	=	SAMP-CoSaMP DAMAS
SNR	=	Signal-to-Noise Ratio
TUHH	=	Technische Universität Hamburg
ZOH	=	Zero-Order Hold

Acknowledgments

I would like to express my sincere gratitude to all those who contributed to the completion of this thesis and made this research journey possible.

First and foremost, I extend my deepest appreciation to Valeo GmbH for providing the opportunity to conduct this research and for their unwavering support throughout this project. The resources, expertise, and collaborative environment provided by Valeo have been instrumental in achieving the objectives of this work.

I am particularly grateful to Michael Klose and Peyman Goli for their invaluable guidance, mentorship, and continuous support. Their technical expertise, insightful discussions, and constructive feedback have significantly enriched this research and contributed to my professional development.

I would also like to thank my academic supervisors for their scholarly guidance, patience, and dedication throughout the research process. Their expertise and commitment to excellence have been essential in shaping this work.

Finally, I express my heartfelt gratitude to my family and friends for their understanding, patience, and moral support throughout this demanding yet rewarding journey. Their belief in my abilities and constant encouragement have been a source of strength and motivation.

This research would not have been possible without the collective support of all these individuals and institutions. Any errors or shortcomings in this work remain solely my responsibility.

1 Abstract

This project develops an advanced simulation framework for sound source localization in complex acoustic environments, addressing the significant challenges present in real-world conditions such as roads and urban settings. Urban acoustic environments present unique challenges for sound localization due to environmental noise [13] [18], complex sound interference patterns, Doppler shifts from moving sources [1] [14], and reverberation effects [2] that can severely degrade signal quality. The simulation framework recreates these phenomena while maintaining precise parameter control, enabling research that would be impractical through field experimentation alone [4].

The proposed system models a configurable sound source with controllable amplitude [3], frequency, and power characteristics, paired with a customizable receiver system implemented as a microphone array. This array can be configured with various microphone types (omnidirectional or directional), each with distinct spatial response patterns requiring proper calibration according to standards like IEC 61094-5 [19]. The simulation incorporates comprehensive signal conditioning to address noise suppression, dynamic range optimization, and frequency filtering, before applying established localization algorithms including Inter Time Difference (ITD), Inter Level Difference (ILD), and beamforming techniques. The framework implements multiple processing approaches including source-time dominant propagation modeling and frequency-domain optimization techniques to balance computational efficiency with acoustic fidelity.

It supports both signal processing approaches: real-time analysis and recording for posterior processing, enabling the creation of diverse training datasets. These datasets can contain either raw waveform data for end-to-end deep neural network (DNN) training or pre-processed features including high-resolution spectrograms, inter-microphone phase differences, time-delay-of-arrival measurements, and spatial cues derived from the array geometry. The simulation accounts for the multifaceted acoustic propagation challenges that make field experiments difficult, including variable environmental conditions (tunnel effects, building reflections, surrounding traffic), multiple acoustic paths, and the physical effects of airflow and temperature variations on sound propagation.

The primary contribution of this research is a holistic simulation-based alternative that generates realistic acoustic scenarios mirroring real-life conditions while remaining modular and extendable. This framework significantly reduces the experimental burden by eliminating the need for costly and time-consuming field deployments that would require extensive infrastructure, specialized equipment, and complex logistics to implement realistic acoustic testing scenarios in urban environments. Additionally, the framework provides a practical alternative to Computational Fluid Dynamics (CFD) approaches, which, while potentially offering higher physical accuracy, demand prohibitive computational resources and processing times—often requiring weeks or months to simulate the diverse acoustic scenarios evaluated in this work. Furthermore, existing CFD solutions typically lack the integrated beamforming algorithms, machine learning capabilities, and comprehensive acoustic signal processing tools that are seam-

lessly incorporated within this unified framework, making this approach uniquely suited for rapid prototyping and extensive parametric studies in acoustic source localization research. By integrating mathematical models from the literature, we’ve constructed a system capable of simulating diverse acoustic environments, creating synthetic datasets, and validating results against physical measurements. This framework serves both as a testbed for developing localization algorithms and as a resource for future researchers who can customize parameters across the entire acoustic pipeline—from environmental acoustics and microphone geometry to signal processing techniques and DNN-based inference—thereby accelerating development in acoustic localization research.

Keywords: reverberant acoustics, Doppler effect, signal filtering, low signal-to-noise ratio, beamforming, sound source localization, source-time dominant approach, deep neural networks

2 Hybrid Acoustic Simulation Framework

This chapter presents the theoretical foundation and implementation details of our proposed hybrid acoustic simulation framework. The framework’s core innovation lies in its integration of source-time dominant propagation modeling with comprehensive reverberation effects, enabling accurate impulse response calculation for complex acoustic systems [2] [4]. Unlike conventional approaches that treat acoustic phenomena in isolation, this framework simultaneously models direct propagation paths, multiple-order reflections from environmental boundaries, motion-induced Doppler effects, and realistic noise characteristics—creating a unified simulation environment that captures the full complexity of real-world acoustic scenarios.

2.1 Recent Advances in Acoustic Source Localization

Now let’s start by examining recent research in acoustic source localization that has made significant progress in addressing specific challenges of complex acoustic environments that we will elaborate on in this chapter, though typically focusing on individual aspects rather than providing a comprehensive solution. This section reviews three key contributions that form the foundation for this thesis.

2.1.1 Literature Review of Recent Approaches

The first approach focuses on reverberation modeling in enclosed spaces. The work by Wen-yong Guo et al. (2023) [2] addresses one of the most persistent challenges in acoustic source localization: the impact of reverberation on static sound sources, particularly in low signal-to-noise ratio conditions. Their research demonstrates that conventional beamforming algorithms often fail in reverberant environments due to their reliance on free-field assumptions. Guo and colleagues specifically focus on reverberant environments such as engine compartment test rooms, where reflections from walls significantly distort the acoustic field, they used room impulse responses that accounts for the specific reflective properties of the environment instead of the traditional free-field Green’s function. Through comparative analysis with conventional beamforming and orthogonal matching pursuit algorithms, they demonstrate that SC-DAMAS achieves superior localization accuracy in reverberant conditions, particularly when the signal-to-noise ratio is low (around 10 dB). Their experiments validate the method’s effectiveness in real-world engine cabin environments, offering a robust solution for static source localization in enclosed spaces.

In contrast, the comprehensive work by Randall Ali and Andrew Christian (2025) [1] focuses on accurately modeling the Doppler shift for moving sound sources. Their research addresses a different but equally important aspect of acoustic localization: the time-varying delay between

a source and receiver that accounts for frequency shifts in moving source scenarios. Ali and Christian identify a critical gap in existing auralization schemes, noting that while time dilation from motion affects signal bandwidth, most current methods inadequately handle the resulting frequency distortions, leading to aliasing and imaging artifacts. Their contribution, the N-Oversampled Transmitted Acoustic Pressure (NOTAP) method, elegantly addresses these artifacts through a combination of oversampling, interpolation, and time-varying filtering. The authors demonstrate how their source-time dominant approach effectively eliminates both aliasing (frequency folding) and imaging (frequency replication) [10] [17]

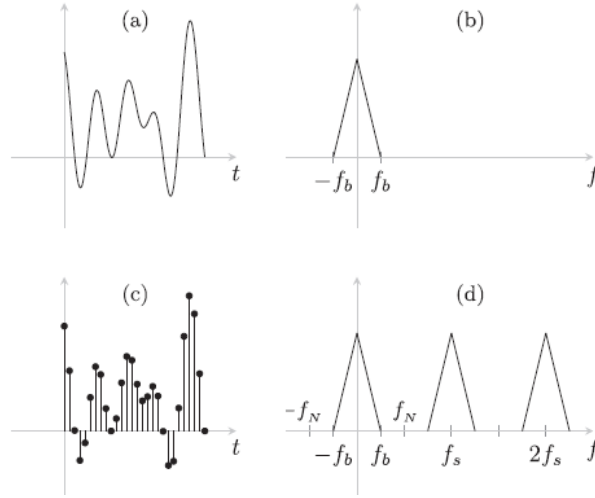


Figure 2.1: The effects of sampling in the time and frequency domain. (a) A continuous time domain signal, (b) its corresponding frequency spectrum with bandwidth f_b , (c) the discrete-time signal after sampling with an impulse train, and (d) the resulting frequency spectrum showing repeated spectral images at multiples of the sampling frequency f_s . Only the first two positive images are shown, but they continue to $\pm\infty$. f_N is the Nyquist frequency. Figure adapted from [1].

Their comparative analysis using spectrograms convincingly demonstrates the superior performance of their approach compared to conventional implementations in maintaining acoustic fidelity despite significant Doppler shifts.

Complementing these specialized approaches, Paolo Chiariotti et al. (2019) [6] provide a comprehensive review of acoustic beamforming techniques [7] [8] for noise source localization. Their work serves as a foundational reference, covering both theoretical concepts and practical applications across diverse acoustic environments. Chiariotti and colleagues categorize and analyze various array designs and beamforming strategies, highlighting how different approaches can be optimized for specific applications. Their review discusses the evolution from traditional delay-and-sum beamforming to more sophisticated techniques like adaptive beamforming, orthogonal beamforming, and deconvolution methods. Particularly relevant to the work is their thorough discussion of microphone array design considerations, including acoustically transparent versus diffractive arrays, and their detailed analysis of performance parameters such as resolution, side-lobe levels, and spatial aliasing. While their overview primarily relies on Green’s function

formulations for free-field conditions, they also address specialized applications like moving source tracking and interior acoustics, providing valuable insights into the practical challenges of implementing beamforming in complex environments.

2.1.2 The Green's Function and Its Limitations

To understand why a hybrid approach is necessary, we must first examine the fundamental limitations of conventional acoustic modeling. Most simulation software packages rely on Green's function [2.1](#) formulation—a mathematical framework that, while elegant in its simplicity, fails to capture the complex acoustic phenomena present in real-world environments. This mathematical construct represents the acoustic pressure response at a receiver point r , caused by a unit point source located at position r_0 , oscillating at an angular frequency ω , within a specified propagation medium such as air, liquid, or solid. Under ideal conditions—specifically, for a homogeneous, lossless, free-field environment without reflective surfaces—the frequency-domain Green's function in three dimensions is given by:

$$G(r, r_0, \omega) = \frac{e^{-jk\|r-r_0\|}}{4\pi\|r-r_0\|} \quad (2.1)$$

In this expression, r and r_0 represent three-dimensional position vectors in space for the receiver and source respectively, with their Euclidean distance $\|r-r_0\|$ determining both amplitude attenuation and phase delay. The angular frequency ω equals $2\pi f$, where f is the frequency in Hertz, while k represents the wavenumber calculated as ω/c , with c being the speed of sound in the medium (approximately 343 m/s in air at standard conditions). The exponential term $e^{-jk\|r-r_0\|}$ accounts for the phase delay as the wave travels through the medium, while the denominator $4\pi\|r-r_0\|$ models the spherical spreading loss that causes amplitude to decrease with distance.

Due to its simplicity, analytical form, and computational efficiency, the Green's function serves as the foundation for many beamforming algorithms, where it functions as the steering vector to predict acoustic pressure at various microphone positions based on assumed or scanned source locations.

However, despite its mathematical elegance and computational efficiency, the Green's function presents significant limitations for complex acoustic scenarios. First, it assumes idealized free-field conditions without accounting for reflections from walls, floors, ceilings, or other obstacles present in real-world environments. Second, it cannot model multiple propagation paths that occur when sound waves bounce off surfaces before reaching the receiver. Third, it fails to capture the Doppler effect and resulting frequency shifts when source positions vary with time, as occurs with moving vehicles, aircraft, or machinery. These limitations make the standard Green's function alone inadequate for acoustic modeling in enclosed spaces, urban environments, or scenarios involving motion.

2.1.3 Proposed Hybrid Approach

To address these limitations, this work integrates and extends the strengths of the aforementioned works. By relying on the Room Impulse Response (RIR) rather than the simple Green's function, the framework provides a more comprehensive model of sound behavior in complex environments. This approach combines the reverberation modeling techniques highlighted by Guo et al., the Doppler shift handling mechanisms developed by Ali and Christian, and the beamforming optimization strategies outlined by Chiariotti et al., offering a more complete solution for acoustic source localization in realistic conditions (Figure 2.2).

Throughout this chapter, theoretical derivation and illustrative case studies demonstrate key concepts and validate theoretical predictions. This pedagogical approach allows immediate visualization of framework capabilities—including impulse response characteristics, frequency domain behavior, energy decay patterns, and RT60 analysis—while comprehensive experimental validation against measured data is presented in Chapter 3. The illustrative examples shown here serve to clarify theoretical concepts and demonstrate practical implementation.

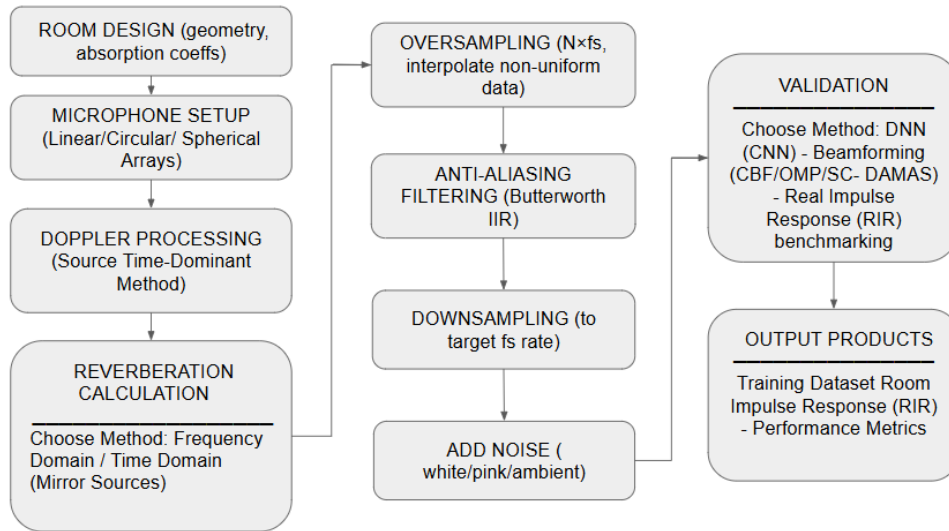


Figure 2.2: provides an overview of the complete implementation pipeline for this hybrid acoustic simulation framework. The following sections detail each component of this workflow.

The framework begins by defining a room setup with specific geometry and physical boundaries, capturing both direct sound paths and multiple reflected paths using the image source method that we will explain in detail for modeling reflections.

For moving source scenarios, the framework computes time-varying delays and frequency shifts, making the system's impulse response a function of time to accurately represent Doppler effects. The model incorporates several signal processing enhancements, including anti-imaging filtering [20], Doppler smoothing, and moving average filters to suppress artifacts and abrupt amplitude changes that might otherwise distort localization results.

To simulate realistic urban conditions, the framework adds low Signal-to-Noise Ratio (SNR) background noise. The model computes impulse responses for point sources and then uses convolution to combine these responses with target signals such as sirens or vehicle noises. For source direction estimation, the framework implements beamforming techniques using multiple processing approaches including source-time dominant propagation modeling and frequency-domain optimization techniques for computational efficiency [15].

Finally, the emission-based microphone signals generated by this model can be used to create datasets for training deep neural networks, which can later be deployed for real-time or offline prediction of source parameters including distance, azimuth, and elevation angles.

By integrating these diverse approaches, this hybrid method overcomes the limitations of individual techniques while leveraging their respective strengths. The following sections will elaborate on each component of this simulation model, demonstrating how they work together.

2.2 Room Acoustics Modeling Implementation

The simulation framework provides configurable acoustic environments designed to replicate urban conditions through adjustable rectangular or cylindrical room geometries with user-defined dimensions. The framework supports optional reverberation modeling using frequency-dependent surface absorption coefficients, multiple ambient noise models (white, pink, and spatially correlated), and optional vehicle body reflections for enhanced realism. Source positioning and trajectory are fully configurable, enabling both stationary and moving source scenarios, while remaining computationally tractable. This spatial configuration allows for the analysis of both near-field and far-field acoustic effects, a critical consideration highlighted by Chiariotti et al. (2019) [6] in their discussion of testing environments for beamforming algorithms.

Within this acoustic space, the framework establishes a defined source trajectory to study motion-induced phenomena. The sound source follows a linear path along the room's Y-axis while maintaining constant X and Z coordinates (Default setup), moving from an initial position toward a final position. This controlled linear motion creates a straightforward velocity vector directed along the Y-axis, simplifying the analysis of Doppler effects while still capturing the essential physics of moving sound sources. The velocity magnitude is derived from the displacement vector and simulation duration through standard kinematic relationships, allowing for parametric adjustment of motion characteristics to test different Doppler conditions.

For acoustic sensing, the framework supports configurable microphone arrays with flexible positioning throughout the acoustic space. The array location and orientation can be systematically adjusted to optimize spatial sampling for different experimental scenarios—with separation distances and geometric arrangements chosen to ensure meaningful signal-to-noise ratios while allowing sufficient spatial development of acoustic phenomena. Following established practices in beamforming research, the framework aligns the coordinate system reference with the array's phase center, which facilitates more intuitive analysis of spatial acoustic patterns regardless of the specific array configuration selected.

This configuration implements the spatial centering condition, where the sum of all microphone position vectors equals zero ($\sum_{m=1}^M \bar{x}_m = 0$). This centering ensures symmetric propagation modeling and provides a mathematically convenient framework for beamforming operations and deep neural network (DNN) source location prediction algorithms. The resulting geometric arrangement creates well-defined direct and reflected acoustic paths that can be precisely modeled through the mirror source implementation, creating an ideal testbed for evaluating reverberation-Doppler interactions and validating localization algorithms.

2.3 Array Geometry Theoretical Framework

2.3.1 Microphone Array Configuration and Implementation

The flexible microphone array system provided here supports multiple geometric configurations: linear, rectangular grid, circular, spiral, and spherical arrangements. This enables comprehensive evaluation of different array topologies and their impact on acoustic source localization performance. The array center position is configurable and can be optimized relative to the source trajectory and room geometry through both interactive and programmatic interfaces.

Each array configuration offers distinct advantages: linear arrays provide computational simplicity and directional sensitivity, rectangular and circular grids provide uniform spatial sampling with known inter-element spacing, facilitating systematic steering vector calculations and predictable beam pattern analysis for conventional beamforming approaches, spiral configurations provide unique co-array properties, and spherical arrangements capture full 3D acoustic fields. The number of microphones is user-configurable, allowing investigation of how sensor density affects localization accuracy across different geometric arrangements.

This multi-configuration approach forms an integral part of this hybrid source-time dominant framework, providing the flexibility to evaluate various array geometries under identical acoustic conditions. The system automatically calculates array quality metrics (F-parameter for spatial unicity) and generates comprehensive acoustic analysis including impulse responses, frequency characteristics, and reverberation parameters for each configuration. As demonstrated in the test-case in this chapter and in Chapter 3, this comparative framework enables systematic analysis of how array geometry influences localization performance under realistic acoustic conditions including Doppler shifts, room reflections, and noise contamination.

2.3.2 Microphone Array Classification and Characteristics

Microphone arrays for acoustic beamforming can be broadly classified into two main categories based on their interaction with the acoustic field. Acoustically transparent arrays, such as those modeled in this framework, are designed so that the microphones, cables, and mounting supports have minimal impact on the sound field they measure. These arrays are particularly

effective for mid to low frequency applications, where the wavelengths are larger than the physical dimensions of the array components. In contrast, acoustically diffractive arrays deliberately incorporate structures (such as rigid spheres) that cause controlled diffraction, altering the sound field in ways that can be exploited for enhanced spatial resolution or directivity.

this implementation uses an acoustically transparent array since the framework focuses primarily on the mathematical modeling of sound propagation rather than the physical interactions between the array structure and the sound field. This approach simplifies the simulation while maintaining fidelity to the core acoustic phenomena being studied.

2.3.3 Array Geometries and Their Performance Characteristics

The performance of microphone arrays is heavily influenced by their geometric arrangement. Geometrical arrays, like for instance, square grid, are characterized by regular patterns of microphone placement. While straightforward to design and implement, these arrays often suffer from aliasing issues and ghost patterns [21] in acoustic maps due to their inherent geometric regularity. Cross-shaped arrays, though simple to build, encounter similar limitations. Circular and multi-circle arrays typically present good performance metrics in terms of figure of merit but still face aliasing challenges from their regular geometry.

To overcome limitations of regular arrays, sparse arrays can be created by strategically removing microphones from lattice structures to create unique inter-sensor spacings, thereby reducing redundancy. However, these arrays may still exhibit residual spurious sidelobes due to remaining spacing periodicity. Random arrays, which feature fully randomized microphone placement, more effectively avoid aliasing problems and reduce systematic patterns in acoustic maps. They can achieve an F parameter of 1 (optimal uniqueness in sensor spacing) [6], though their Maximum Sidelobe Level (MSL) performance may not be optimal when using a limited number of sensors.

Spiral arrays, including Archimedean, Dougherty log-spiral, Arcondoulis spiral, and multi-arm spiral configurations, offer several advantages for advanced applications. These arrays can cover large apertures with relatively few sensors while maintaining low MSL across a wide frequency range. Their good circular symmetry, zero-redundancy, and broad base of inter-microphone vector spacing make them particularly valuable for aeroacoustic applications. For instance, an Underbrink spiral array with approximately 100 sensors can achieve a dynamic range of 12 dB over a frequency range of 1–80 kHz.

For applications requiring performance across widely separated frequency ranges, nested arrays combine a small array (optimized for high frequencies) and a large array (optimized for low frequencies) within the same structure. This approach enables effective tuning of array size and shape for specific frequency ranges, improving the frequency ratio from 10:1 up to 100:1. The sensors in these arrays can be either independent or shared between the two component arrays.

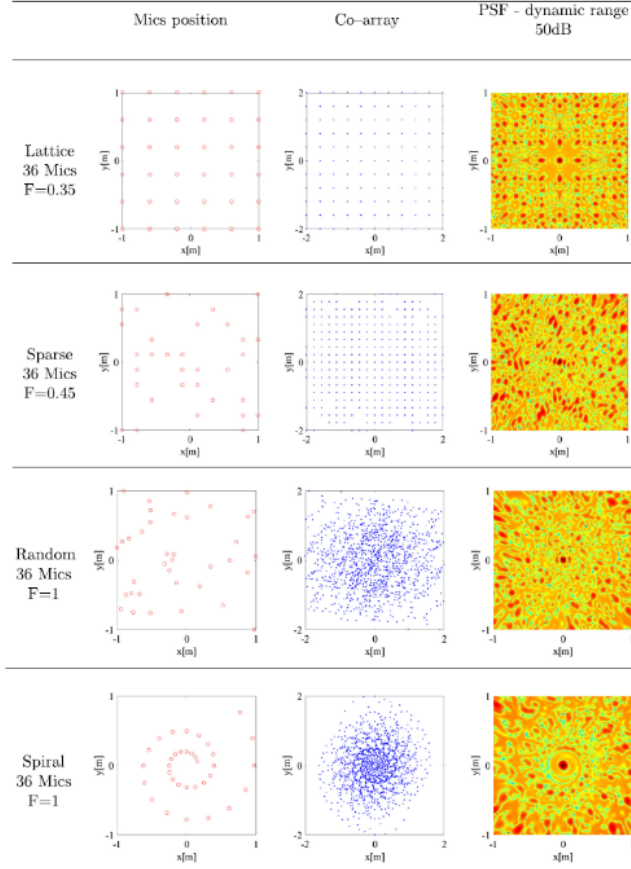


Figure 2.3: Example Comparison of microphone array geometries and their corresponding co-array structures and point spread functions (PSF). Four different array configurations are shown. Left column shows microphone positions in spatial coordinates, middle column displays the co-array structure representing all pairwise microphone separations, and right column presents the point spread function with 50dB dynamic range illustrating the array's spatial resolution characteristics. . [6]

2.3.4 Performance Considerations for Array Design

There are also several key factors influence array performance and should be considered during design. The inter-element spacing represents the most critical design parameter in microphone array configuration, governing frequency response and preventing spatial aliasing artifacts that can completely compromise localization accuracy. The fundamental constraint requires that inter-element spacing d satisfies $d < \lambda/2 = c/(2f)$, where λ is the acoustic wavelength, c is sound speed, and f is frequency. When this condition is violated ($d > \lambda/2$), grating lobes appear in the beam pattern, creating false source directions that can lead to catastrophic localization errors.

This wavelength criterion becomes increasingly challenging at higher frequencies where λ decreases, requiring tighter sensor spacing to maintain aliasing-free operation. The number of

microphones determines the array’s ability to resolve closely spaced sources and suppress interference, while the spatial arrangement affects the array’s unicity properties (discussed later through the F -parameter analysis). Proper inter-element spacing design is therefore the foundational requirement for reliable acoustic source localization across the intended frequency range. Circular symmetry is often desirable to avoid azimuthal dependency in resolution mapping, while non-redundancy in sensor spacing helps reduce spatial aliasing problems.

The design optimization process involves careful balancing of performance requirements, with trade-offs between resolution, Maximum Sidelobe Level, operating frequency range, and the practical constraint of sensor count. The F parameter, which measures the percentage of uniqueness in sensor spacing, serves as a useful figure of merit for evaluating array quality during this optimization process.

Given the demonstrated impact of array geometry on localization performance, future extensions should incorporate more specialized geometries such as reflective spherical arrays, where microphones are flush-mounted on rigid spherical shells. Such diffractive configurations would act as acoustic lenses, creating enhanced phase differences at low frequencies while providing directional masking effects at higher frequencies. With this comprehensive understanding of each geometry’s strengths and limitations, the choice of array geometry should ultimately be guided by specific application requirements, including frequency range of interest, expected source distribution, environmental characteristics, and practical constraints such as channel count and installation feasibility.

2.4 Sound Source Modeling and Physical Phenomena

2.4.1 Monopole Sound Source Characteristics and Implementation

Having established the array geometry framework, we now turn to the acoustic source modeling. This simulation framework models a monopole point source as the fundamental acoustic excitation model. Monopole sources represent the simplest form of acoustic radiator, characterized by omnidirectional spherical radiation patterns where sound propagates uniformly in all directions from a single point. Monopole sources serve as essential building blocks for acoustic modeling due to their mathematical tractability and ability to represent compact acoustic sources.

The simulation framework incorporates multiple source signal types including siren sounds, linear chirp signals, and exponential sine sweeps. Among these options, the exponential sine sweep provides optimal characteristics as a reference signal due to its superior characteristics for acoustic system identification and impulse response measurement. Exponential sweeps provide several key [22] advantages: (1) optimal spectral energy distribution across the frequency range of interest, (2) excellent signal-to-noise ratio enabling robust deconvolution for impulse response extraction, (3) controlled spectral characteristics that facilitate accurate Doppler shift analysis, and (4) compatibility with standard room acoustic measurement protocols, allowing direct comparison with experimental data.

monopole sources are characterized by precisely controlled acoustic properties, which provides sufficient energy for detection while remaining computationally manageable. The omnidirectional nature of monopole radiation ensures that the propagation patterns are governed purely by the room acoustics and not by source directivity, thereby isolating the effects of the environment on sound propagation. This approach aligns with the methodology employed by Guo et al [2], who similarly used monopole sources when validating their beamforming algorithms for reverberant environments.

One significant advantage of using monopole sources is their well-defined mathematical representation through the Green's function. As Ali et al [1] note in their work on Doppler modeling, the analytical tractability of monopoles makes them ideal for validating more complex acoustic processing algorithms. Additionally, monopoles represent a reasonable approximation for many real-world compact noise sources such as small machinery components or localized leaks, making them relevant for practical applications.

The model initially processes the monopole sound and after successfully calculating the system's impulse response, the framework expands this approach by convolving this impulse response with a recorded siren sound to simulate real-world emergency vehicle scenarios. This convolution step transforms the idealized monopole radiation into a more complex, realistic acoustic signature while maintaining the physical accuracy of the propagation model 2.8. The convolution process effectively demonstrates how any arbitrary sound source would be modified by the simulated acoustic environment, incorporating both direct sound transmission and multi-path reflections from room boundaries.

2.4.2 Sound Propagation in Reverberant Environments

When sound propagates in enclosed spaces, reflections from surrounding surfaces create a complex acoustic field distinctly different from free-field propagation. This phenomenon is readily observable in everyday scenarios, such as when driving through a tunnel as a police or ambulance siren passes by—the sound becomes noticeably fuller, with extended reverberation that causes the siren to persist and echo long after the vehicle has passed, creating a characteristic 'tail' of reflected sound that gradually fades away. In these reverberant environments, sound waves don't simply travel directly from source to receiver but instead follow multiple paths, including reflections from walls, ceilings, and other surfaces. The resulting sound field at any measurement point becomes a superposition of the direct signal and numerous delayed reflections, each arriving with different amplitudes and phase shifts.

The received acoustic signal $y(t)$ 2.2 in such conditions can be represented mathematically through the convolution of the source signal with the room's impulse response function:

$$y(t) = \int h(\tau) x(t - \tau) d\tau + n(t) = h(t) * x(t) + n(t) \quad (2.2)$$

Here, $h(t)$ 2.4 represents the room impulse response (RIR) function that characterizes the acoustic environment, $x(t)$ is the source signal, and $n(t)$ accounts for ambient noise present in the system. The delay variable τ represents the time differences introduced by various reflection

paths, while the convolution operator $(*)$ captures how the source signal is modified by the room acoustics.

This time-domain representation can also be expressed in the frequency domain 2.3 as:

$$Y(\omega) = X(\omega)H(\omega) + N(\omega) \quad (2.3)$$

This formulation highlights the crucial role of the indoor impulse response function $h(t)$, which serves as the essential bridge between the emitted signal and what is ultimately received at the sensor array. Accurate computation of this impulse response is therefore fundamental to any realistic simulation of reverberant acoustic environments.

2.4.3 Mirror Source Method

The challenge lies in accurately computing all the delayed reflections that comprise the room impulse response $h(t)$. To address this computational complexity, this framework employs the mirror source method, an approach that provides an elegant solution for modeling reflected sound waves in enclosed spaces. Its core concept involves replacing reflections with virtual “mirror” sources positioned symmetrically across reflecting surfaces. This mathematical transformation allows reflections to be computed as if they were additional direct sound paths from these virtual sources, significantly simplifying the simulation while maintaining physical accuracy.

Table 2.1 presents the mirror source reflection path computation algorithm. The reflection coefficients β are calculated from absorption coefficients γ using the relationship $\beta = \sqrt{1 - \gamma}$, converting energy ratios to amplitude ratios since acoustic energy is proportional to the square of pressure amplitude.

For each reflecting surface in the room, a mirror image of the original sound source is created by reflecting the source position across that surface. Higher-order reflections (where sound bounces off multiple surfaces before reaching the receiver) are modeled by creating mirror images of mirror images. Each reflection’s amplitude is attenuated according to the reflection coefficients of the surfaces it interacts with, accounting for the energy lost during each bounce.

The resulting room impulse response based on the mirror method can be formulated as:

$$h(t) = \sum \sum \beta_{x1}^{|n_x - q|} \beta_{x2}^{|n_x|} \beta_{y1}^{|n_y - j|} \beta_{y2}^{|n_y|} \beta_{z1}^{|n_z - k|} \beta_{z2}^{|n_z|} \cdot \frac{\delta\left(t - \frac{|\mathbf{R}_p + \mathbf{R}_r|}{c}\right)}{4\pi|\mathbf{R}_p + \mathbf{R}_r|} \quad (2.4)$$

In this expression, the reflection coefficients β_{x1} , β_{y1} , and β_{z1} represent the acoustic properties of walls near the coordinate origin, while β_{x2} , β_{y2} , and β_{z2} correspond to opposite walls. The integer set (n_x, n_y, n_z) denotes the reflection order along each spatial axis. The vector $\mathbf{R}_p$

Table 2.1: Algorithm 2 for Mirror Source Reflection Path Computation

1:	Input the room dimensions $roomDim = [L_x, L_y, L_z]$, maximum reflection order n_{max} , and absorption coefficients $\gamma_{x1}, \gamma_{x2}, \gamma_{y1}, \gamma_{y2}, \gamma_{z1}, \gamma_{z2}$
2:	Calculate reflection coefficients: $\beta = \sqrt{1 - \gamma}$ for each wall
3:	Initialize reflection path counter $path_counter = 1$ and empty reflection paths structure
4:	for $n_x = -n_{max}$ to n_{max} do
5:	for $n_y = -n_{max}$ to n_{max} do
6:	for $n_z = -n_{max}$ to n_{max} do
7:	if $n_x = 0$ and $n_y = 0$ and $n_z = 0$ then continue
8:	if $ n_x + n_y + n_z > n_{max}$ then continue
9:	Calculate reflection coefficients:
10:	$\beta_x = \begin{cases} \beta_{x1}^{ n_x } & \text{if } n_x < 0 \\ \beta_{x2}^{ n_x } & \text{if } n_x > 0 \\ 1.0 & \text{if } n_x = 0 \end{cases}$
11:	$\beta_y = \begin{cases} \beta_{y1}^{ n_y } & \text{if } n_y < 0 \\ \beta_{y2}^{ n_y } & \text{if } n_y > 0 \\ 1.0 & \text{if } n_y = 0 \end{cases}$
12:	$\beta_z = \begin{cases} \beta_{z1}^{ n_z } & \text{if } n_z < 0 \\ \beta_{z2}^{ n_z } & \text{if } n_z > 0 \\ 1.0 & \text{if } n_z = 0 \end{cases}$
13:	$\beta_{total} = \beta_x \cdot \beta_y \cdot \beta_z$
14:	Calculate spatial offset: $offset = [2n_x L_x, 2n_y L_y, 2n_z L_z]$
15:	Store path: $reflection_paths(path_counter) = \{[n_x, n_y, n_z], \beta_{total}, offset\}$
16:	$path_counter = path_counter + 1$
17:	end for
18:	end for
19:	end for
20:	return $reflection_paths$ with total count $path_counter - 1$
Parameters description: $\gamma \rightarrow$ Absorption coefficients, $\beta \rightarrow$ Reflection coefficients, $n_x, n_y, n_z \rightarrow$ Reflection indices along spatial axes, $offset \rightarrow$ Mirror source spatial displacement	

represents the distance from the actual sound source or mirror image to the reflection point, while $\mathbf{R}_r$ corresponds to the virtual space size for multi-order reflections. The speed of sound c and the Dirac delta function δ ensure proper timing of energy arrival at the receiver position.

The mirror source method effectively handles the geometric aspects of reflections, but implementing realistic room simulation requires addressing a fundamental physical complexity: the frequency-dependent behavior of reflections. Unlike the simplified case where reflection co-

efficients remain constant, accurate modeling requires evaluating reflection coefficients across multiple frequency bands, typically ranging from 125 Hz to 4000 Hz [2]. This is necessary because wall materials exhibit nonlinear acoustic responses where absorption varies significantly with frequency. In this simulation framework, reflection coefficients β are calculated from absorption coefficients γ using the standard relationship as shown in table 2.2

Table 2.2: Sound absorption coefficient of each wall of the room

Frequency (Hz)	125	250	500	1000	2000	4000
Front and left side	0.18	0.06	0.04	0.03	0.02	0.02
Right side and ceiling	0.10	0.05	0.06	0.07	0.09	0.08
Ground	0.01	0.01	0.01	0.01	0.02	0.02
Back side	0.35	0.25	0.18	0.12	0.07	0.04

This approach reveals that wall reflection varies substantially across the frequency spectrum with higher frequencies generally reflecting more effectively while lower frequencies are more readily absorbed, depending on the material properties. For instance, a wall with absorption coefficients [0.18, 0.06, 0.04, 0.03, 0.02, 0.02] across the frequency spectrum absorbs 18% of sound energy at 125 Hz but only 2% at 4000 Hz, indicating increased reflectivity at higher frequencies—a characteristic behavior of many common building materials.

For computational efficiency, the framework employs the 1000 Hz reflection coefficients as representative values across the full frequency spectrum. While the source signal contains broadband spectral content, this simplification significantly reduces computational overhead while preserving realistic spatial and temporal reverberation characteristics. The 1000 Hz frequency band is particularly appropriate as it falls within the midrange of typical acoustic sources and coincides with the frequency range where human auditory perception is most sensitive. This approach maintains the full bandwidth of the source signal while applying uniform reflection attenuation based on the midband coefficients.

The reflection path computation employs nested iterations across three spatial dimensions (n_x, n_y, n_z) , where each index represents the number of reflections along that axis. For a maximum reflection order n_{max} , the algorithm evaluates all combinations where $|n_x| + |n_y| + |n_z| \leq n_{max}$, efficiently identifying valid reflection paths while excluding computationally expensive higher-order reflections that contribute minimally to the overall acoustic field.

The framework uses positive values for reflection coefficients because they represent the ratio of reflected sound amplitude to incident sound amplitude. Using negative reflection coefficients would physically represent inverting the phase of the reflected wave, which is not the intended behavior in this general room acoustics model. These pre-computed reflection paths store geometric indices, combined reflection coefficients, and spatial offsets, with dynamic effects such as Doppler shifts applied separately during the signal propagation stage when source motion is present.

By incorporating these frequency-dependent reflection characteristics, this simulation achieves greater fidelity when computing the room impulse response, particularly when dealing with

broadband acoustic sources like moving vehicles, speech, or sirens. This frequency-aware approach significantly enhances the accuracy of downstream processing tasks such as beamforming and acoustic localization.

2.4.4 The Doppler Effect and Time-Varying Delays

When a sound source moves relative to a receiver, the perceived frequency of the sound changes due to the Doppler effect. This phenomenon results from the continuously changing distance between source and receiver, which creates a time-varying propagation delay. Understanding and accurately modeling this effect is essential for realistic acoustic simulation of moving sources.

The mathematical relationship between the emission time t_e (when sound leaves the source) and the receiver time t_r (when it arrives at the microphone) for a moving source and stationary receiver is governed by the following nonlinear equation 2.5:

$$t_e + \frac{|\mathbf{r}_s(t_e) - \mathbf{r}_r|}{c} = t_r \quad (2.5)$$

Here, $\mathbf{r}_s(t_e)$ represents the position of the sound source at emission time t_e , $\mathbf{r}_r$ is the fixed position of the receiver, c is the speed of sound in the medium, and $|\cdot|$ denotes the Euclidean distance between the positions. This equation captures the physical reality that sound emitted at time t_e travels at speed c over the distance between source and receiver positions, arriving at time t_r .

Solving this equation presents a significant computational challenge because the source position $\mathbf{r}_s(t_e)$ is itself a function of the unknown emission time t_e . This creates a nonlinear relationship that typically requires iterative numerical methods such as Newton-Raphson [28] iterations or MATLAB's `fzero` function to solve for t_e at each discrete receiver time t_r . The complexity increases when dealing with arbitrary motion patterns or when combining the Doppler effect with reverberation modeling.

The framework addresses this computational challenge through multiple approaches. The source-time dominant method avoids the nonlinear equation entirely by calculating forward propagation using uniformly sampled source times t_s to compute corresponding receiver times via:

$$t_r = t_s + \frac{|\mathbf{r}_s(t_s) - \mathbf{r}_r|}{c} \quad (2.6)$$

This forward propagation approach [1] aligns naturally with the physical reality of acoustic wave propagation, eliminating the need for iterative root-finding algorithms while maintaining physical accuracy. For applications requiring receiver-time synchronization, iterative methods

can solve the nonlinear relationship. Frequency-domain techniques provide additional computational optimization by processing Doppler effects in the spectral domain.

The accurate resolution of this time-varying relationship is crucial for capturing several important acoustic phenomena:

- Time dilation effects, where the temporal pattern of the source signal becomes stretched or compressed at the receiver
- Frequency shifts that vary throughout the source trajectory, particularly as it approaches or recedes from the receiver, automatically accounting for Doppler effects through the time-varying source positions as shown in the Doppler factor computation 2.16
- Variations in arrival time that affect the perceived rhythm or timing of acoustic events

These effects fundamentally alter how sounds are perceived and interpreted, making their accurate simulation essential for realistic acoustic modeling of moving sources in both free-field and reverberant environments.

The source-time dominant propagation process involves four core computational stages: transmission time calculation, oversampling and interpolation, anti aliasing [5] filtering, and temporal resampling. For each uniformly sampled source emission time, the framework calculates when the acoustic energy reaches each receiver position.

The resulting non-uniformly sampled receiver timeline is then processed through oversampling techniques to prevent aliasing artifacts. Anti-aliasing filters remove spectral images introduced by interpolation, a process that will be examined in detail in the following section, while final resampling produces uniformly sampled output signals suitable for analysis and processing.

This approach directly addresses the computational challenges of moving source simulations while maintaining high acoustic fidelity.

2.5 source-time dominant formulation

In the source-time dominant formulation, the framework begins by defining a vector of uniformly sampled source emission times 2.7:

$$t_s = [t_s(0), t_s(1), \dots, t_s(M-1)]^T \quad (2.7)$$

For each source emission time sample $t_s(k)$, the framework computes the corresponding receiver arrival time $t_r(k)$ using the direct forward propagation relationship 2.8:

$$t_r(k) = t_s(k) + \frac{\|\mathbf{r}_s(t_s(k)) - \mathbf{r}_r\|}{c} \quad (2.8)$$

Where $\mathbf{r}_s(t_s(k))$ represents the position of the moving sound source at emission time $t_s(k)$, $\mathbf{r}_r$ is the fixed position of the microphone, and c is the speed of sound in air. This equation captures the physical reality that sound emitted at time t_s travels at speed c over the distance between source and receiver positions, arriving at the calculated time t_r .

Since the source position is explicitly known at each emission time. The source signal amplitude at emission time is retrieved directly 2.9:

$$p_s(k) = p_{source}(t_s(k)) \quad (2.9)$$

The propagation distance 2.10 between source and receiver at each emission instant is computed as:

$$r(k) = \|\mathbf{r}_s(t_s(k)) - \mathbf{r}_r\| \quad (2.10)$$

This distance calculation serves multiple purposes: amplitude scaling for attenuation effects, Doppler factor computation, and propagation delay determination. For amplitude scaling, the framework supports an enhanced model that incorporates both linear and quadratic attenuation terms 2.13:

$$a_{geometric}(r) = \frac{1}{4\pi r^2} \quad (2.11)$$

$$a_{atmospheric}(r) = e^{-\alpha r} \quad (2.12)$$

$$a_{total}(r) = a_{geometric}(r) \cdot a_{atmospheric}(r) = \frac{e^{-\alpha r}}{4\pi r^2} \quad (2.13)$$

The amplitude attenuation model incorporates two fundamental physical mechanisms that govern sound propagation in air. The geometric spreading term (Equation 2.11) represents the natural decrease in sound intensity due to the expansion of acoustic wavefronts, following the inverse square law. This effect is purely geometric and frequency-independent, reflecting the conservation of acoustic energy as it spreads over increasingly larger areas with distance from the source. The atmospheric absorption term (Equation 2.12) accounts for the molecular absorption of acoustic energy by air molecules, where α represents the frequency-dependent absorption coefficient. This exponential decay becomes more pronounced at higher frequencies and is influenced by atmospheric conditions including temperature, humidity, and pressure. The combined effect (Equation 2.13) provides a physically accurate representation of sound attenuation, ensuring that the simulation adheres to established principles of acoustic propagation while maintaining computational efficiency.

To accurately model the Doppler effect, the framework can use two approaches. the rate of change in receiver arrival times as shown in equation (2.16), or directly from the relative velocity as shown in 2.14:

$$D = \frac{c}{c - v_{rel}}, \quad v_{rel} = \vec{v} \cdot \frac{\vec{r}_s - \vec{r}_r}{|\vec{r}_s - \vec{r}_r|} \quad (2.14)$$

where $\vec{v}$ is the source velocity vector and v_{rel} is the radial component of velocity relative to the receiver. This velocity-based approach can offer improved numerical stability in certain scenarios.

To prevent discontinuities near closest approach points, smoothing is applied:

$$D_{smooth}(i) = \frac{1}{2W + 1} \sum_{j=i-W}^{i+W} D(j) \quad (2.15)$$

where W is the smoothing window half-width, typically 5-10 samples depending on the motion characteristics.

However, the source-time dominant approach enables a more direct and computationally efficient method by computing the instantaneous Doppler factor directly from the time dilation effects in the receiver timeline 2.16:

$$D(k) = \frac{f_s}{1/(t_r(k+1) - t_r(k))} \quad (2.16)$$

This time-domain approach offers several advantages: (1) it avoids numerical differentiation errors inherent in velocity estimation, (2) it naturally captures the time dilation effects of forward propagation, and (3) it maintains numerical stability near closest approach points where velocity-based methods can become unstable.

The forward-propagated signals create non-uniformly sampled receiver time series due to Doppler-induced time dilation. To address this, first, the framework creates an oversampled receiver time grid and applies zero-order hold interpolation to map the irregularly spaced arrival times to uniform sampling. The oversampled signal $p_{over}(j)$ is then processed through anti-aliasing filtering and downsampled to produce the final uniformly sampled output.

2.5.1 Oversampling Grid Creation

The framework creates a fine temporal grid to accommodate the non-uniform arrival times resulting from Doppler-induced time dilation. The oversampled temporal grid is defined as:

$$t_{over} = \left\{ t_{min} + \frac{k}{N \cdot f_s} \mid k \in \mathbb{Z}, t_{min} \leq t_{min} + \frac{k}{N \cdot f_s} \leq t_{max} \right\} \quad (2.17)$$

where $t_{min} = \min(t_r)$ and $t_{max} = \max(t_r)$ represent the temporal bounds of the non-uniformly spaced receiver arrival times, N is the oversampling ratio (typically 8-16), and f_s is the target sampling frequency. This creates a uniform temporal grid with sufficient resolution to capture the irregularly spaced acoustic arrivals without introducing temporal quantization errors.

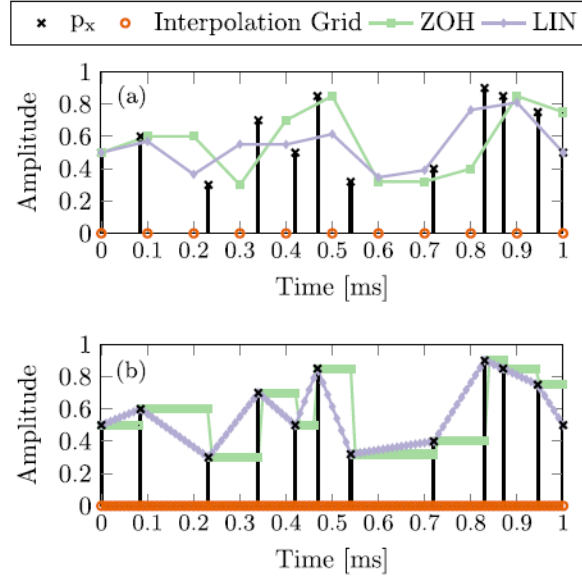


Figure 2.4: Different levels of oversampling with different simple interpolation schemes, where p_x is the transmitted acoustic pressure with nonuniformly spaced samples, i.e., the input signal for the interpolation schemes. (a) Undersampled receiver, where aliasing is possible. (b) Highly sampled receiver, producing a large frequency range above the possible transmitted content where aliasing is possible. [1]

2.5.2 Zero-Order Hold Interpolation with Physical Attenuation

The source signal values are mapped to the oversampled grid using zero-order hold interpolation combined with physically accurate distance-based attenuation. For each oversampled time instant $t_{over}(j)$, the corresponding signal value is determined by:

$$p_{over}(j) = p_s(i) \cdot a_{total}(r(i)) \quad (2.18)$$

where $p_{over}(j)$ represents the acoustic pressure at the receiver location on the oversampled temporal grid, effectively creating a high-resolution representation of the irregularly spaced

acoustic arrivals. This oversampled signal p_{over} serves as an intermediate processing stage that captures the non-uniform timing of Doppler-affected arrivals while providing sufficient temporal resolution to prevent aliasing during subsequent filtering operations.

The zero-order hold interpolation maintains the amplitude of each source sample $p_s(i)$ over the time interval until the next arrival, as illustrated in Figure 2.4. This approach maps each oversampled time point to the most recent source emission that would have arrived by that time, implementing the causal nature of acoustic propagation. While the complete attenuation model $a_{total}(r(i))$ from Equation 2.13 ensures physical accuracy in amplitude scaling based on propagation distance, so that p_{over} represents the actual acoustic pressure magnitude expected at the receiver position accounting for both geometric spreading and atmospheric absorption effects.

2.5.3 Preventing Aliasing and Imaging

Aliasing and Imaging can significantly degrade audio quality and localization accuracy if left unchecked [5] [3].

Aliasing occurs when the receiver’s sampling frequency is insufficient to capture the Doppler-shifted bandwidth of the source signal. This is particularly problematic for sources moving toward the receiver, as frequencies are compressed into higher bands, potentially violating the Nyquist-Shannon sampling theorem. Even a stationary signal that originally lies within the Nyquist limit can have portions of its spectral energy pushed into aliased territory through Doppler compression, corrupting the time-frequency representation.

Imaging, by contrast, arises primarily from the interpolation of non-uniformly sampled data—an inevitable consequence when reconstructing signals affected by Doppler-induced time dilation. When basic interpolation methods like zero-order hold (ZOH) or linear interpolation are applied to irregularly spaced samples, spectral replicas or “images” (as shown in figure 2.1) can appear in the reconstructed waveform. These artifacts become especially problematic when downsampling without adequate anti-imaging filtering, leading to baseband contamination and perceptual distortion.

To address aliasing, the framework introduces an oversampling stage at a frequency:

$$f_{s,o} = N \cdot f_{s,r} \quad (2.19)$$

Where $f_{s,o}$ is the oversampling frequency, $f_{s,r}$ is the target receiver sampling frequency, and N is the oversampling ratio (typically $N \gg 1$). This creates a finer temporal grid for interpolating the acoustic pressures, allowing even heavily Doppler-compressed signals to be reconstructed without violating the Nyquist criterion.

The oversampled signal contains both the desired baseband content and unwanted image bands introduced by interpolation. To eliminate these images, a 6th-order Butterworth IIR lowpass

filter is applied with a fixed cutoff frequency, effectively removing frequency components above the receiver's Nyquist limit, including those introduced by Doppler shifting.

The filtered output is computed using the difference equation:

$$p_{filtered}(l) = \sum_{k=0}^6 b_k \cdot p_{over}(l - k) - \sum_{k=1}^6 a_k \cdot p_{filtered}(l - k) \quad (2.20)$$

where $p_{over}(l)$ represents the input signal (oversampled data), $p_{filtered}(l)$ is the filtered output signal, b_k are the feedforward coefficients, and a_k are the feedback coefficients with $a_0 = 1$.

To design this Butterworth filter, the coefficients are derived from the analog prototype with magnitude-squared response:

$$|H_a(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{12}} \quad (2.21)$$

where ω_c is the analog cutoff frequency. Converting this analog design to digital implementation requires the bilinear transformation (Tustin's method):

$$s = \frac{2}{T} \cdot \frac{z - 1}{z + 1} \quad (2.22)$$

where $s = j\omega$ is the Laplace domain variable, T is the sampling period, and z is the z-domain variable. The bilinear transformation introduces frequency warping through the nonlinear relationship

$$\omega = \frac{2}{T} \tan\left(\frac{\Omega T}{2}\right),$$

where the entire analog frequency range (0 to ∞) maps to the digital range (0 to π/T). While this warping affects frequency scaling, particularly at higher frequencies, Tustin's method is preferred over alternatives like impulse invariant (which avoids warping but introduces aliasing) or matched z-transform (which maps only specific frequencies exactly) because it preserves filter stability and prevents aliasing artifacts.

To compensate for frequency warping effects, the cutoff frequency is strategically set below the Nyquist limit:

$$f_c = 0.8 \cdot \frac{f_s}{2} \quad (2.23)$$

For the oversampled processing domain, the normalized digital cutoff frequency becomes:

$$\omega_{c,\text{norm}} = \frac{f_c}{f_{s,\text{over}}/2} = \frac{0.8 \cdot f_s}{f_{s,\text{over}}} \quad (2.24)$$

The final digital transfer function preserves the characteristic Butterworth magnitude response:

$$|H(e^{j\omega})|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_{c,\text{norm}}}\right)^{12}} \quad (2.25)$$

This design ensures maximally flat passband response with -3dB attenuation at the cutoff frequency and monotonic rolloff in the stopband. The digital transfer function $H(z)$ is then decomposed into its rational polynomial form:

$$H(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^6 b_k z^{-k}}{1 + \sum_{k=1}^6 a_k z^{-k}} \quad (2.26)$$

where the numerator polynomial $B(z)$ yields the feedforward coefficients b_k and the denominator polynomial $A(z)$ provides the feedback coefficients a_k . This rational decomposition enables direct implementation of the difference equation, providing effective anti-aliasing while maintaining computational efficiency through the recursive IIR structure.

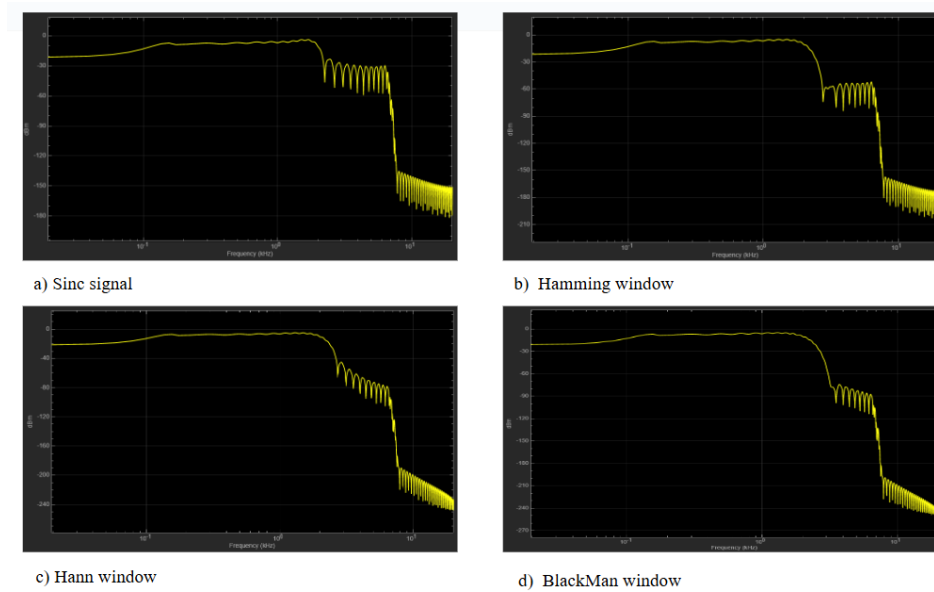


Figure 2.5: FIR windowing comparison showing precise linear-phase characteristics and guaranteed stability. This external comparison illustrates why FIR approaches were initially considered during framework design, but this framework ultimately adopted IIR filters 2.6 for superior computational efficiency in real-time acoustic processing applications.

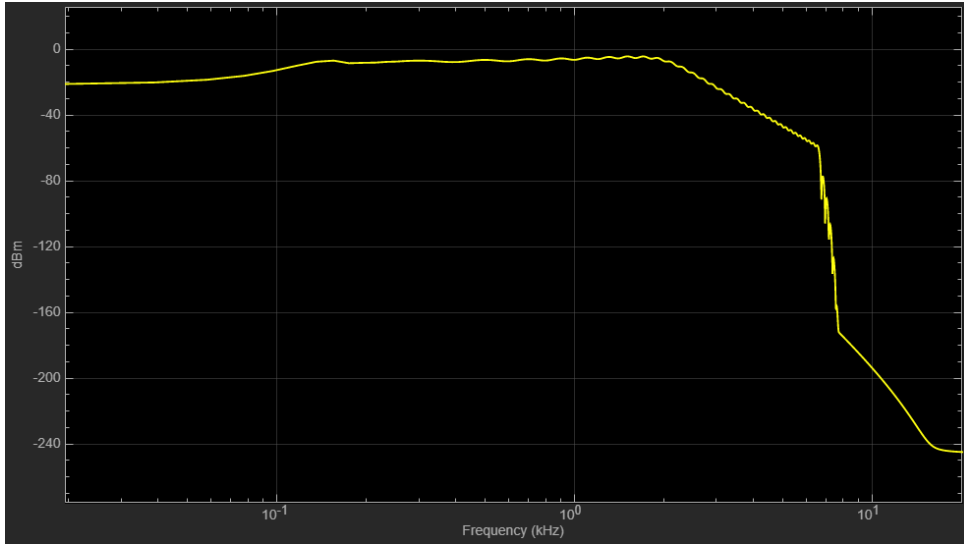


Figure 2.6: Butterworth IIR filter response showing smooth monotonic rolloff and superior stopband attenuation without sidelobe artifacts. This external filter analysis demonstrates why this implementation utilizes the IIR approach due to its computational efficiency and excellent frequency response characteristics, accepting the trade-off of phase distortion for practical real-time performance.

As demonstrated in Figure 2.6, the Butterworth design provides the optimal balance between computational efficiency and frequency response characteristics required for this real-time acoustic processing framework. By utilizing this fixed Butterworth filter design, this source-time dominant approach successfully prevents both aliasing and imaging artifacts, ensuring high-fidelity acoustic reconstructions for subsequent localization and analysis tasks.

2.5.4 Downsampling

With the spectral artifacts removed by filtering, the framework returns to the target sampling rate through systematic decimation. The downsampling operation selects every N -th sample from the filtered oversampled signal, where N corresponds to the oversampling ratio used in the initial grid creation:

$$p_{direct}(k) = p_{filtered}(N \cdot k + 1), \quad k = 0, 1, 2, \dots, \left\lfloor \frac{L_{filtered} - 1}{N} \right\rfloor \quad (2.27)$$

where $L_{filtered}$ is the length of the filtered signal and N is the oversampling ratio. This decimation process systematically "jumps" through the filtered data, retaining samples at indices $\{1, N + 1, 2N + 1, 3N + 1, \dots\}$ while discarding the intermediate samples that provided the temporal resolution necessary for artifact-free filtering. The result is a signal sampled at the original target rate f_s with all Doppler-induced temporal distortions properly reconstructed and spectral artifacts eliminated.

2.6 Reflection Processing Implementation Strategies

Table 2.3: Algorithm 3 for Reflection Processing Implementation Strategies

1: Input reflection paths from mirror source method, source signal $x(t)$, processing method choice 2: if frequency_domain_method then 3: Compute source FFT: $X(f) = \mathcal{F}\{x(t)\}$ 4: for each reflection path j do 5: Apply transfer function: $Y_j(f) = X(f) \cdot \beta_{total,j} \cdot e^{-j2\pi f\tau_j}$ 6: end for 7: Combine paths: $Y_{total}(f) = \sum_{j=1}^{N_{paths}} Y_j(f)$ 8: Return time signal: $y(t) = \mathcal{F}^{-1}\{Y_{total}(f)\}$ 9: else if time_domain_method then 10: for each reflection path j do 11: Compute mirror trajectory: $\mathbf{s}_j(t) = \mathbf{s}(t) + \mathbf{offset}_j$ 12: Calculate distances: $d_j(t_s) = \ \mathbf{s}_j(t_s) - \mathbf{r}\ _2$ 13: Determine arrival times: $t_{r,j}(t_s) = t_s + d_j(t_s)/c$ 14: if optimized_processing then 15: Create oversampled grid at $f_{over} = N_{oversample} \cdot f_s$ 16: Apply ZOH interpolation across full signal 17: else if chunked_processing then 18: Divide timeline into chunks of size N_{chunk} 19: for each chunk m do 20: Apply ZOH mapping within chunk 21: end for 22: end if 23: Reconstruct signal: $p_j(t) = p_s(k) \cdot \beta_{total,j} \cdot \frac{1}{4\pi d_j^2} \cdot e^{-\alpha d_j}$ 24: end for 25: Combine all reflection paths 26: end if 27: return processed reflection signals Parameters: $\beta_{total,j} \rightarrow$ Reflection coefficients, $\tau_j \rightarrow$ Propagation delays, $N_{chunk} \rightarrow$ Chunk size (typically 2000), $N_{oversample} \rightarrow$ Oversampling ratio
--

The framework implements three distinct computational approaches for processing acoustic reflections, each optimized for different performance requirements and computational constraints. These methods utilize the pre-computed reflection paths from the mirror source method (Section 2.4.3) but differ fundamentally in their signal processing implementation:

1. **Frequency-Domain Processing:** Processes all reflection paths simultaneously through spectral manipulation, offering maximum computational efficiency through FFT operations and vectorized processing.
2. **Chunked Time-Domain Processing:** Divides the temporal processing into memory-efficient segments while maintaining time-domain accuracy, balancing computational feasibility with physical fidelity.
3. **Optimized Time-Domain Processing:** Employs full oversampling across the entire signal duration for maximum temporal accuracy, though at significant memory cost.

Each approach implements the same underlying physics through the source-time dominant formulation established in Section 2.3, but trades off computational efficiency, memory requirements, and temporal accuracy differently.

2.6.1 Frequency-Domain Processing

The frequency-domain approach processes reflections through spectral manipulation, offering computational efficiency through vectorized FFT operations. The source signal is first transformed to the frequency domain:

$$X(f) = \mathcal{F}\{x(t)\} \quad (2.28)$$

where $X(f)$ represents the frequency spectrum of the source signal $x(t)$, and $\mathcal{F}\{\cdot\}$ denotes the Fourier transform operation.

Each reflection path is then processed using frequency-domain transfer functions that incorporate both the reflection characteristics and propagation delays:

$$Y_j(f) = X(f) \cdot \beta_{total,j} \cdot e^{-j2\pi f\tau_j} \quad (2.29)$$

where $Y_j(f)$ is the frequency response for reflection path j , $\beta_{total,j}$ represents the combined reflection coefficient (analogous to the same coefficient used in the time-domain formulation), and $e^{-j2\pi f\tau_j}$ implements the propagation delay τ_j as a phase shift in the frequency domain. This approach achieves the same physical modeling as the time-domain methods but through spectral manipulation rather than temporal convolution.

The contributions from all reflection paths are combined through vectorized summation, and the final time-domain signal is recovered through inverse transformation:

$$Y_{total}(f) = \sum_{j=1}^{N_{paths}} Y_j(f), \quad y(t) = \mathcal{F}^{-1}\{Y_{total}(f)\} \quad (2.30)$$

where $Y_{total}(f)$ represents the complete frequency-domain response including all reflection paths, and $y(t)$ is the final time-domain signal containing all reflected acoustic components. This vectorized approach processes all reflection paths simultaneously, providing significant computational advantages over sequential time-domain processing.

2.6.2 Time-Domain Physical Formulation

Both time-domain approaches employ the same rigorous physical formulation based on mirror source trajectories, implementing the source-time dominant methodology established in Section 2.3. This formulation provides exact temporal accuracy by tracking acoustic propagation through explicit geometric calculations.

For each reflection path j , the mirror source trajectory is computed by applying the spatial transformation derived from the mirror source method:

$$\mathbf{s}_j(t) = \mathbf{s}(t) + \mathbf{offset}_j \quad (2.31)$$

where $\mathbf{s}_j(t)$ represents the position of the mirror source at time t , $\mathbf{s}(t)$ is the actual source trajectory, and $\mathbf{offset}_j$ represents the spatial offset vector derived from the mirror source method that accounts for reflections off room boundaries.

The acoustic propagation for each reflection path requires computing the instantaneous distance from mirror source to receiver, followed by the corresponding arrival time calculation:

$$d_j(t_s) = \|\mathbf{s}_j(t_s) - \mathbf{r}\|_2, \quad t_{r,j}(t_s) = t_s + \frac{d_j(t_s)}{c} \quad (2.32)$$

where $d_j(t_s)$ is the Euclidean distance between the mirror source position at emission time t_s and the receiver position $\mathbf{r}$, $t_{r,j}(t_s)$ represents the arrival time at the receiver for sound emitted at time t_s , and c is the speed of sound. This formulation captures the fundamental physics of acoustic propagation while accounting for the extended path lengths introduced by reflections.

The signal reconstruction incorporates comprehensive physical attenuation effects, combining reflection losses with propagation-based amplitude scaling:

$$p_j(t) = p_s(k) \cdot \beta_{total,j} \cdot \frac{1}{4\pi d_j^2(t_s)} \cdot e^{-\alpha d_j(t_s)} \quad (2.33)$$

where $p_j(t)$ is the reconstructed acoustic pressure for reflection path j at receiver time t , $p_s(k)$ represents the source signal amplitude at the appropriate emission time, $\beta_{total,j}$ is the combined reflection coefficient (identical to that used in the frequency-domain approach), $\frac{1}{4\pi d_j^2(t_s)}$ implements geometric spreading losses, and $e^{-\alpha d_j(t_s)}$ accounts for atmospheric absorption. The index $k = \arg \max_n \{t_{r,j}(n) \leq t\}$ represents the most recent source emission that would have arrived

by receiver time t , implementing the causal nature of acoustic propagation through zero-order hold interpolation.

2.6.3 Time-Domain Processing Approaches

The two time-domain implementations differ only in their computational strategy while maintaining identical physical accuracy through the formulation established above.

Optimized Time-Domain Processing. This approach applies the complete physical formulation from Equations 2.31–2.33 using full oversampling at frequency $f_{over} = N_{oversample} \cdot f_s$ across the entire signal duration. The oversampling factor $N_{oversample}$ (typically $10\times$) provides maximum temporal resolution for capturing rapid Doppler variations and ensuring artifact-free signal reconstruction, delivering the highest possible temporal accuracy at the cost of substantial memory requirements.

Chunked Time-Domain Processing. To address the memory limitations of the optimized approach while preserving physical fidelity, the chunked method processes the receiver timeline in manageable segments. The receiver time vector is divided into chunks of size N_{chunk} (typically 2000 samples):

$$t_{chunk,m} = [t_r(m \cdot N_{chunk}), \dots, t_r((m+1) \cdot N_{chunk} - 1)] \quad (2.34)$$

where $t_{chunk,m}$ represents the temporal segment for chunk m , enabling processing of long signals without memory overflow. The same rigorous physical formulation (Equations 2.31–2.33) is applied within each chunk, maintaining identical acoustic accuracy to the optimized method while constraining memory usage to practical levels. This approach successfully balances computational feasibility with the temporal precision required for accurate Doppler modeling and reflection processing.

2.6.4 Performance Comparison and Memory Scaling

The memory requirements scale differently across approaches:

$$\begin{aligned} \text{Memory}_{frequency} &= \mathcal{O}(N_{total} \times N_{paths}) \\ \text{Memory}_{chunked} &= \mathcal{O}(N_{chunk} \times N_{paths}) \\ \text{Memory}_{optimized} &= \mathcal{O}(N_{total} \times N_{paths} \times N_{oversample}) \end{aligned} \quad (2.35)$$

Empirical performance results for a test scenario (Doppler effects, first-order reflections, 4 microphones) demonstrate significant differences:

Table 2.4: Reflection Processing Performance Comparison

Method	Time per Mic	Memory per Mic	Efficiency (paths/sec)	Status
Chunked Time-Domain	8.19s	12.9 MB	0.7	Success
Optimized Time-Domain	–	17.2 GB+	–	Memory Error
Frequency Domain	0.414s	4.4 MB	14.5	Success

The frequency-domain approach demonstrates significant computational advantages: **20× faster processing** (0.414s vs. 8.19s), **3× lower memory usage** (4.4 MB vs. 12.9 MB), and **20× higher efficiency** (14.5 vs. 0.7 paths/sec). These advantages stem from FFT computational efficiency, elimination of large oversampled arrays, and natural vectorization of frequency-dependent reflection processing.

However, the frequency-domain approach sacrifices temporal accuracy through averaged Doppler approximations and spectral bin quantization effects. The time-domain chunked approach maintains higher physical fidelity by preserving continuous Doppler variations and exact temporal mapping, making it preferable for applications requiring precise acoustic modeling despite the computational overhead.

The selection between approaches should be guided by application priorities: frequency-domain processing for computational efficiency in preliminary design or interactive simulations, and time-domain processing for high-fidelity acoustic analysis requiring accurate Doppler representation and temporal precision.

2.7 Noise Characterization in Source-Time Dominant Processing

2.7.1 Environmental Noise Integration

The framework supports multiple environmental noise models applied after all acoustic processing is complete, including white noise with uniform spectral density, pink noise with 1/f spectral characteristics, and spatially correlated ambient noise with 70% common base component and 30% microphone-specific variations.

All noise models maintain a fixed 10 dB SNR relative to individual microphone signal power, providing consistent testing conditions across different scenarios. This SNR selection provides moderate testing conditions suitable for algorithm development and validation. While urban acoustic monitoring studies document significantly more challenging conditions in real-world environments, Research has documented emergency vehicle sirens operating at SNRs as low as 2.2 dB in urban scenarios [25], while studies of emergency vehicle detection systems report the need for robust performance at SNRs of -15 dB or below for practical deployment [26]. Additionally, comprehensive urban acoustic databases show SNR levels ranging from -10 dB to +15 dB with high heterogeneity of intermediate SNR levels [27], indicating that this 10 dB

selection provides moderate yet realistic testing conditions without the extreme challenges that would require sophisticated preprocessing techniques such as spectral subtraction or adaptive filtering before localization processing.

The modular noise architecture enables straightforward integration of more sophisticated environmental models. The noise generation is applied individually to each microphone using:

$$p_{noisy,m}(t) = p_{clean,m}(t) + \sqrt{\frac{P_{signal,m}}{10^{SNR_{dB}/10}}} \cdot n_m(t) \quad (2.36)$$

where $p_{noisy,m}(t)$ is the final noisy signal at microphone m , $p_{clean,m}(t)$ represents the clean acoustic signal after all processing, $P_{signal,m}$ is the signal power for microphone m , $SNR_{dB} = 10$ dB is the target signal-to-noise ratio, and $n_m(t)$ is the noise signal (white, pink, or spatially correlated ambient noise). This per-microphone implementation ensures that each array element receives appropriately scaled noise relative to its individual signal power, maintaining consistent SNR conditions across the entire microphone array while preserving realistic spatial noise characteristics.

2.8 Impulse Response Extraction via Deconvolution

Following the signal processing pipeline described in the previous sections, this framework implements a deconvolution technique to extract room impulse responses from both simulated and measured microphone signals. This process is fundamental to this approach as it separates the acoustic environment's characteristics from the source signal properties, enabling precise analysis of the room's acoustic behavior independently of the excitation signal used.

The deconvolution process serves dual purposes within this framework: first, it extracts impulse responses from simulated signals for analysis and validation; second, it processes real measurement data for comparison with simulation results as detailed in the experimental validation. This unified approach ensures consistency between the simulation framework and real-world validation procedures, enabling direct quantitative comparison between theoretical predictions and empirical observations.

The deconvolution process employs regularized frequency-domain division to solve the fundamental relationship between the recorded signal, source signal, and room impulse response. Given a recorded signal $y(t)$ and known source signal $x(t)$, the room impulse response $h(t)$ can be extracted through the frequency-domain relationship:

$$Y(\omega) = X(\omega) \cdot H(\omega) + N(\omega) \quad (2.37)$$

where $Y(\omega)$, $X(\omega)$, and $H(\omega)$ are the Fourier transforms of the recorded signal, source signal, and impulse response respectively, and $N(\omega)$ represents additive noise in the frequency domain.

Table 2.5: Algorithm 4 for Template-Aligned Impulse Response Extraction and Comparison

1: Input recorded signal data from measurement system, simulated signal data from acoustic framework 2: Generate reference exponential sine sweep: $x_{ref}(t) = \sin(2\pi K(e^{t/L} - 1))$ 3: Extract distinctive template from reference signal middle region (0.9s duration) 4: for each signal type (recorded, simulated) do 5: Perform FFT-based normalized cross-correlation with template 6: Find optimal alignment: $\tau_{optimal} = \arg \max_{\tau} R_{norm}[\tau] $ 7: Extract aligned segment based on template match position 8: end for 9: Verify alignment quality through correlation analysis 10: for each aligned signal do 11: Zero-pad to length $N = 2^{\lceil \log_2(L_y + L_x - 1) \rceil}$ 12: Compute FFTs: $X(\omega) = \mathcal{F}\{x_{ref}\}$, $Y(\omega) = \mathcal{F}\{y_{aligned}\}$ 13: Apply regularized deconvolution: $H(\omega) = \frac{Y(\omega) \cdot X^*(\omega)}{ X(\omega) ^2 + \lambda \cdot \max(X(\omega) ^2)}$ 14: Extract impulse response: $h(t) = \mathcal{F}^{-1}\{H(\omega)\}$ 15: Apply post-processing: causality enforcement, fade-out, noise suppression 16: end for 17: Perform comprehensive comparison: time domain, frequency domain, acoustic metrics 18: Calculate statistical validation metrics: correlation, NRMSE, SNR 19: Generate RT60, clarity indices, and early-to-late energy ratios 20: return extracted impulse responses and comprehensive comparison results Parameters: $K, L \rightarrow$ Sweep parameters, $\lambda \rightarrow$ Regularization factor, $R_{norm} \rightarrow$ Normalized cross-correlation, $\tau \rightarrow$ Temporal lag
--

A direct attempt to solve for $H(\omega)$ via division would yield:

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \quad (2.38)$$

However, this approach becomes numerically unstable when $X(\omega) \rightarrow 0$ at certain frequencies, leading to noise amplification and spurious artifacts in the extracted impulse response. To address this fundamental limitation, the framework implements regularized deconvolution using an enhanced Wiener filter approach:

$$H(\omega) = \frac{Y(\omega) \cdot X^*(\omega)}{|X(\omega)|^2 + \lambda(\omega) \cdot \max(|X(\omega)|^2)} \quad (2.39)$$

where $X^*(\omega)$ denotes the complex conjugate of $X(\omega)$, $|X(\omega)|^2$ represents the power spectral density of the source signal, and $\lambda(\omega)$ is a frequency-dependent regularization parameter that provides adaptive stabilization.

2.8.1 Template-Based Signal Alignment for Simulation Validation

A critical challenge in validating acoustic simulation frameworks lies in the temporal alignment of recorded and simulated signals for meaningful comparison. While both signals contain identical source content—in this case, exponential sine sweeps spanning 20 Hz to 20 kHz—their temporal positioning within the recording differs significantly, as real-world acoustic measurements introduce inherent timing uncertainties that complicate direct comparison with simulation results. These uncertainties arise from several sources: manual triggering delays in measurement systems, variable processing latencies in recording equipment, and the arbitrary selection of simulation time origins. Additionally, recorded signals often contain pre-signal silence, post-signal reverberation, and background noise that extend beyond the core acoustic content of interest.

In contrast, the simulation framework generates signals with precise temporal control, typically starting the acoustic source at $t = 0$ and maintaining deterministic timing throughout the propagation modeling process. To address these alignment challenges, the framework implements a template-based signal alignment methodology that automatically identifies and aligns the corresponding acoustic content regardless of absolute timing differences. This approach enables robust comparison between simulated and measured impulse responses extracted through deconvolution.

Reference Template Generation. The template matching process begins with the generation of a distinctive reference template derived from the known source signal characteristics. The framework extracts a 900-millisecond template from the middle portion of the exponential sine sweep, specifically targeting the frequency range where the sweep exhibits maximum energy density and spectral uniqueness—typically the 500-2000 Hz region where room acoustic effects are most pronounced and signal-to-noise ratios are optimal.

The template selection algorithm identifies the most energetic and spectrally distinctive portion of the reference signal:

$$T(t) = S_{ref}(t_{start} : t_{start} + T_{duration}) \quad (2.40)$$

where $T(t)$ represents the extracted template, $S_{ref}(t)$ is the reference exponential sine sweep, t_{start} corresponds to approximately 30% into the sweep duration (avoiding fade-in artifacts), and $T_{duration} = 0.9$ seconds provides sufficient temporal resolution for reliable matching.

Normalized Cross-Correlation Matching. The template matching employs FFT-based normalized cross-correlation to efficiently locate the best temporal alignment between the template and both recorded and simulated signals. This approach provides computational efficiency while maintaining robust performance under varying noise conditions.

For each target signal $x(t)$, the normalized cross-correlation is computed as:

$$R_{norm}[\tau] = \frac{\sum_t T(t) \cdot x(t + \tau)}{\sqrt{\sum_t T(t)^2} \sqrt{\sum_t x(t + \tau)^2}} \quad (2.41)$$

where τ represents the temporal lag, and the normalization ensures correlation values remain bounded between -1 and +1, providing a reliable similarity metric independent of signal amplitude differences.

The optimal alignment position is determined by:

$$\tau_{optimal} = \arg \max_{\tau} |R_{norm}[\tau]| \quad (2.42)$$

This approach automatically compensates for the manual triggering delays inherent in laboratory measurements, enabling precise temporal alignment **based on acoustic content rather than absolute timing references**.

Signal Extraction and Verification. Once optimal alignment positions are identified for both recorded and simulated signals, corresponding temporal segments are extracted using identical window lengths. The extraction process ensures both signals maintain the same temporal duration and sampling rate, facilitating subsequent deconvolution and comparison analyses.

Alignment quality is verified through direct correlation analysis of the extracted segments:

$$\rho_{verification} = \frac{COV(x_{recorded}, x_{simulated})}{\sigma_{recorded} \sigma_{simulated}} \quad (2.43)$$

where $\rho_{verification}$ provides a quantitative measure of alignment success. Correlation coefficients above 0.3 typically indicate successful alignment, while lower values may suggest insufficient signal-to-noise ratios or fundamental differences in acoustic content.

Integration with Deconvolution Pipeline

The regularized deconvolution process successfully extracts physically meaningful impulse responses as demonstrated in Figure 2.8, where the template-matched results show clear spatial variations across the 4-microphone linear array (as in the Test Case Analysis - section 2.9) with distinct acoustic characteristics at each sensor location. This integrated approach ensures that validation comparisons reflect genuine differences in acoustic modeling rather than artificial temporal misalignment artifacts, validating both the template alignment methodology and the deconvolution accuracy.

The deconvolution process applies identical regularization parameters and processing techniques to both aligned signals:

$$H_{extracted}(\omega) = \frac{Y(\omega) \cdot X^*(\omega)}{|X(\omega)|^2 + \lambda \cdot \max(|X(\omega)|^2)} \quad (2.44)$$

where $Y(\omega)$ represents the FFT of the aligned recorded or simulated signal, $X(\omega)$ is the reference sweep spectrum, and λ provides frequency-dependent regularization for numerical stability.

Validation Metrics and Analysis Framework

The template-based alignment methodology enables comprehensive validation analysis across multiple acoustic domains:

Time Domain Validation. Cross-correlation analysis quantifies temporal similarity between extracted impulse responses, providing insight into the framework’s ability to capture onset timing, early reflection patterns, and decay characteristics. The alignment process compensates for measurement timing uncertainties while preserving relative temporal relationships essential for acoustic accuracy assessment.

As demonstrated in Figure 3.3, the template-aligned impulse responses exhibit coherent onset patterns and energy decay characteristics that validate the accuracy of the temporal synchronization approach.

Frequency Domain Validation. Spectral comparison of aligned signals reveals the simulation’s fidelity in reproducing frequency-dependent room transfer characteristics, modal behavior, and absorption effects. The template matching ensures that frequency domain comparisons reflect genuine acoustic modeling accuracy rather than temporal misalignment artifacts. 3.4 illustrates the preserved spectral envelope structures achieved through template alignment, enabling meaningful frequency-domain validation despite systematic amplitude differences.

Acoustic Metrics Validation. Standard room acoustic parameters including RT60, clarity indices (C50/C80), and early-to-late energy ratios can be reliably computed from template-aligned impulse responses, providing quantitative validation metrics that directly assess the framework’s acoustic modeling capabilities. The signal decomposition analysis shown in Figure 3.8 demonstrates the framework’s ability to separate and validate direct path, reflection, and combined acoustic components through template-based alignment.

This template-based validation methodology establishes a robust foundation for the experimental validation studies presented in Chapter 3, ensuring that comparisons between simulated and measured acoustic responses provide meaningful insights into simulation framework accuracy and limitations.

Adaptive Regularization Strategy

The regularization parameter λ plays a critical role in balancing noise suppression and frequency response accuracy. this implementation employs a multi-tier adaptive strategy that considers both signal characteristics and spectral quality. The base regularization is defined as:

$$\lambda = \alpha \cdot \beta_{\text{SF}} \cdot \max(|X(\omega)|^2) \quad (2.45)$$

where α is a signal-dependent scaling factor and β_{SF} is a spectral flatness correction factor. The signal-dependent factor α is determined according to signal type and spectral coverage:

$$\alpha = \begin{cases} 0.001 & \text{for exponential sine sweeps (high spectral uniformity)} \\ 0.002 & \text{for linear chirps (moderate spectral uniformity)} \\ 0.005\text{--}0.01 & \text{for siren signals (poor spectral coverage)} \\ 0.001 \cdot (2 - \text{SF}) & \text{for adaptive spectral-based scaling} \end{cases} \quad (2.46)$$

The spectral flatness correction factor adapts regularization strength based on local spectral characteristics:

$$\beta_{\text{SF}} = 1 + \frac{(1 - \text{SF})^2}{4} \quad (2.47)$$

This quadratic relationship ensures that signals with poor spectral uniformity receive progressively stronger regularization, effectively mitigating noise amplification in frequency regions with low excitation energy.

Implementation Details and Artifact Prevention

Zero-Padding for Circular Convolution Prevention. To avoid circular convolution artifacts inherent in FFT-based processing, both signals are zero-padded to ensure linear convolution behavior. The optimal padding length N is computed as:

$$N = 2^{\lceil \log_2(L_y + L_x - 1) \rceil} \quad (2.48)$$

where L_y and L_x are the lengths of the recorded and source signals respectively, and $\lceil \cdot \rceil$ denotes the ceiling function. This ensures that the convolution output length $(L_y + L_x - 1)$ fits within the next power-of-two for efficient FFT computation while preventing time-domain wraparound.

Enhanced Spectral Flatness Analysis. this implementation employs a sliding-window spectral flatness measure to provide localized spectral quality assessment:

$$\text{SF}(\omega_c) = \frac{\left(\prod_{i=c-W/2}^{c+W/2} |X(\omega_i)| \right)^{1/W}}{\frac{1}{W} \sum_{i=c-W/2}^{c+W/2} |X(\omega_i)|} \quad (2.49)$$

where ω_c is the center frequency of the analysis window, W is the window size (typically 64 frequency bins), and the analysis is performed across overlapping windows spanning the signal bandwidth. SF values approaching unity indicate uniform spectral distribution, while values near zero indicate tonal or sparse spectral content requiring stronger regularization.

Multi-Stage Post-Processing Pipeline. The extracted impulse response undergoes comprehensive post-processing to ensure physical validity and remove extraction artifacts:

$$h_{\text{final}}(n) = w_{\text{causal}}(n) \cdot w_{\text{fade}}(n) \cdot w_{\text{denoise}}(n) \cdot h_{\text{raw}}(n) \quad (2.50)$$

where: - $w_{\text{causal}}(n)$ enforces causality by zeroing pre-trigger content - $w_{\text{fade}}(n)$ applies exponential fade-out over the final 50 ms: $w_{\text{fade}}(n) = \exp(-\tau(n - n_{\text{fade}}))$ for $n > n_{\text{fade}}$ - $w_{\text{denoise}}(n)$ applies noise floor suppression based on late reverberation analysis

Quality Assessment and Validation Metrics. The deconvolution process incorporates real-time quality assessment through multiple metrics:

$$\text{Coherence}(\omega) = \frac{|G_{xy}(\omega)|^2}{G_{xx}(\omega) \cdot G_{yy}(\omega)} \quad (2.51)$$

$$\text{Crest Factor} = \frac{\max(|h(n)|)}{\text{RMS}(h(n))} \quad (2.52)$$

$$\text{Dynamic Range} = 20 \log_{10} \left(\frac{\max(|h(n)|)}{\text{noise floor}} \right) \quad (2.53)$$

Coherence values below 0.8 in critical frequency bands trigger automatic re-processing with adjusted regularization parameters.

Integration with Simulation Framework

The deconvolution module integrates seamlessly as a unified post-processing stage within the source-time dominant framework. This integration ensures consistent processing across both simulated and measured acoustic data, maintaining methodological coherence throughout the analysis pipeline. The processing sequence follows the established computational paradigm. After computing all acoustic phenomena—including direct paths, mirror source reflections, and environmental noise—the aggregated signals undergo temporal resampling to establish uniform sampling grids. The regularized deconvolution then extracts impulse responses using the enhanced Wiener filtering approach described previously. Quality assessment metrics continuously monitor extraction fidelity, automatically triggering parameter adjustment when coherence values fall below acceptable thresholds. This systematic integration maintains consistency between simulation and measurement processing, ensuring that extracted impulse responses reflect comparable analytical conditions.

2.9 Framework Performance Demonstration: Test Case Analysis

To demonstrate the theoretical concepts developed in the previous sections, this section presents illustrative examples using a representative 4-microphone linear array configuration in a controlled acoustic environment. This section showcases the framework’s capabilities through systematic analysis of impulse response characteristics, frequency domain behavior, energy decay patterns, and reverberation analysis without added environmental noise, providing clear visualization of the underlying acoustic phenomena before comprehensive experimental validation in Chapter 3.

2.9.1 Implementation Configuration

The demonstration employs a standardized test configuration with a $4 \times 4 \times 4$ meter rectangular room and stationary monopole source at $[2, 2, 2]$ meters, generating an exponential sine sweep (20 Hz to 20 kHz). The 4-microphone linear array, positioned at $[0.5, 0.5, 0.5]$ meters, features 10 cm inter-element spacing. Processing uses 3rd-order reflections (62 paths), 16 kHz sampling, $10\times$ oversampling, and regularized deconvolution with $\lambda = 0.001$. Linear arrays are selected for this demonstration due to their prevalence in automotive acoustic applications, where they can be readily integrated into vehicle body panels, and their computational efficiency for validating the fundamental framework capabilities before scaling to more complex multi-array configurations.

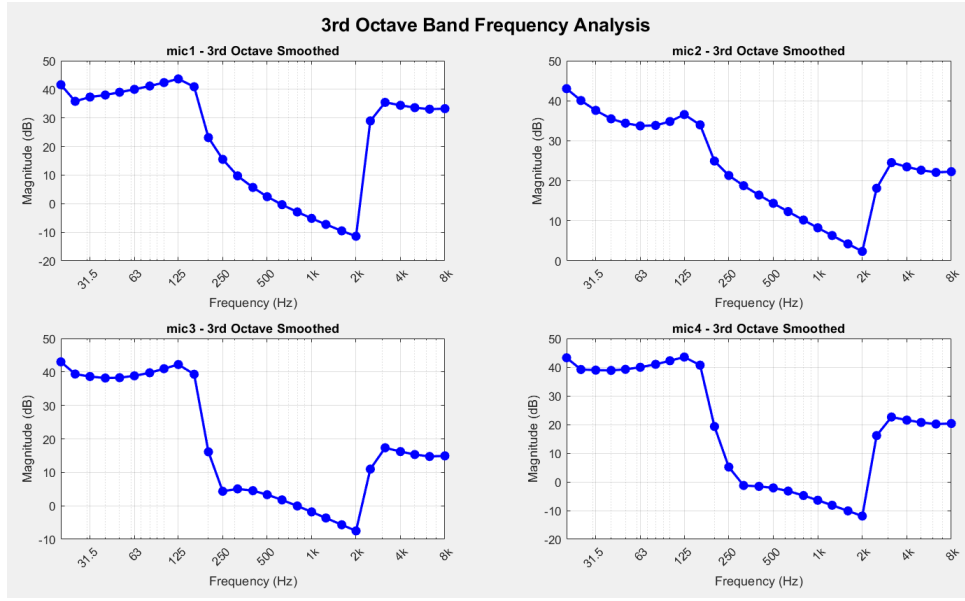


Figure 2.7: 3rd octave band smoothed frequency analysis of the 4-microphone linear array impulse responses.

2.9.2 Frequency Domain Analysis

Figure 2.7 shows characteristic room acoustic behavior with strong low-frequency response peaking around 31.5-63 Hz (40-50 dB), deep modal nulls in the mid-frequency range around 250-500 Hz (dropping to -10 dB), recovery in the 1-2 kHz region, and high-frequency rolloff above 4 kHz. Each microphone exhibits position-dependent spectral variations confirming accurate spatial modeling.

2.9.3 Impulse Response Characteristics

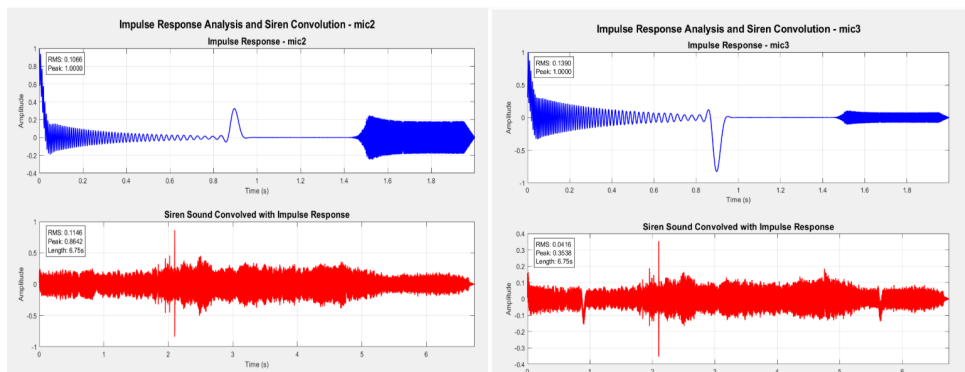


Figure 2.8: Impulse response analysis and siren convolution demonstration showing simulated room acoustics effects.

The impulse responses in Figure 2.8 demonstrate characteristic exponential decay with onset around 1-2 ms and reverberation extending to approximately 4 seconds. Spatial variations between microphones reflect geometric relationships to source and room boundaries, with microphone 2 showing a more compact impulse response compared to microphone 3's extended decay pattern. The convolution with siren signals validates realistic acoustic modification through room transfer functions, demonstrating how the extracted impulse responses effectively process complex acoustic signals.

2.9.4 Energy Decay and Multi-Channel Coherence

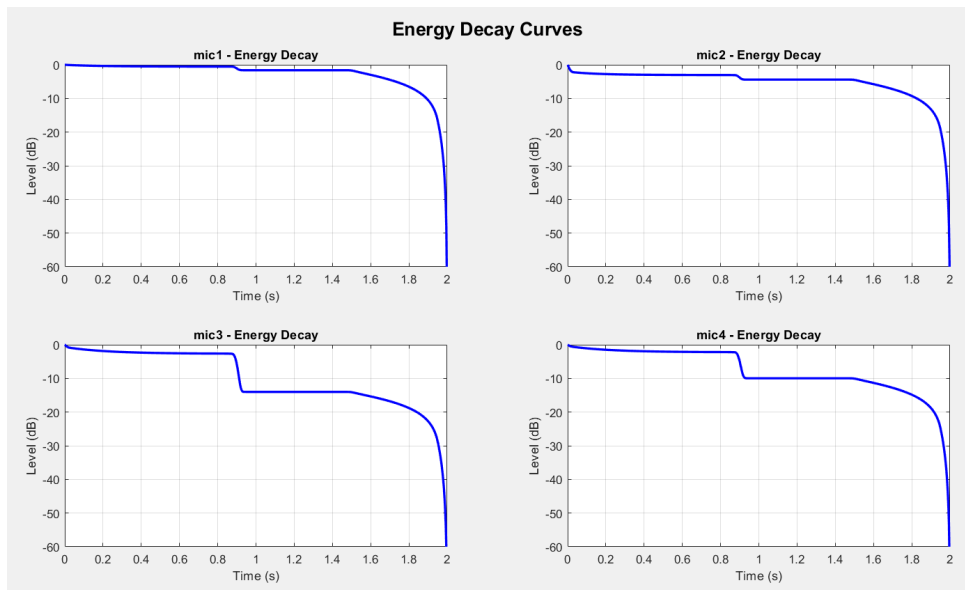


Figure 2.9: Energy decay curves for four-microphone linear array showing reverberation characteristics.

Figure 2.9 illustrates varying exponential decay patterns across microphones, with steep energy decay extending over approximately 2 seconds to reach -60 dB levels. Microphone 3 exhibits a distinctive sharp transition around 1 second, while others show smooth decay characteristics. These spatial variations validate position-dependent acoustic modeling in the reverberant field.

2.9.5 Reverberation Time Analysis

The RT60 analysis in Figure 2.10 reveals frequency-dependent reverberation behavior consistent with room acoustic theory. The template-matched impulse responses show normalized characteristics with corresponding RMS and peak values for each microphone, demonstrating the framework's ability to capture realistic decay patterns across the frequency spectrum.

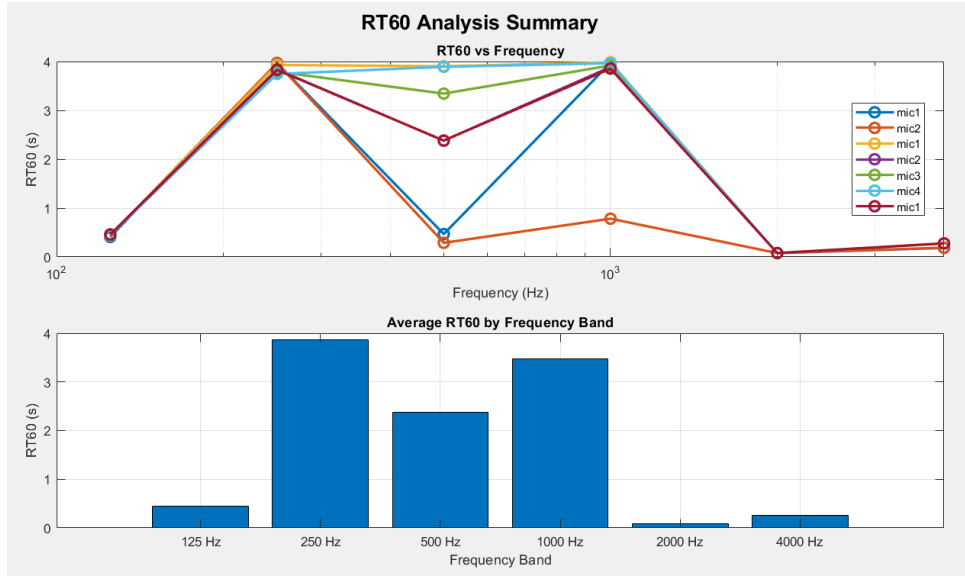


Figure 2.10: RT60 analysis summary for 4-microphone array showing frequency-dependent reverberation characteristics.

2.9.6 Detailed Impulse Response Extraction

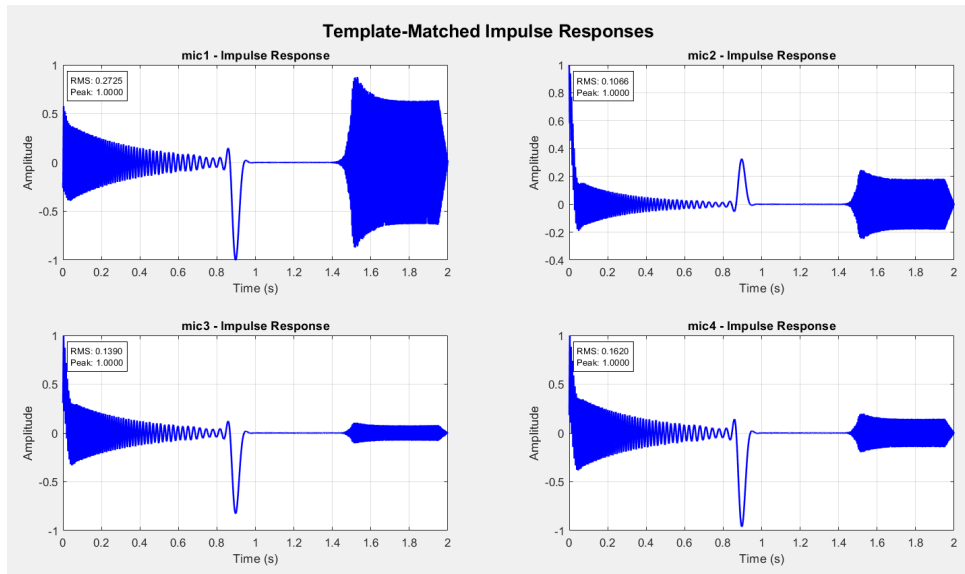


Figure 2.11: Template-matched impulse responses showing deconvolved room characteristics across the linear array.

Figure 2.11 presents normalized impulse responses with clear spatial variations across the array. Distinct onset timing around 0.6-0.8 seconds, varying amplitude characteristics, and exponential decay patterns reflect geometric relationships between source and microphone positions. The template-based alignment enables direct comparison of acoustic characteristics, validating microphone-specific propagation effects.

2.9.7 Signal Processing Validation

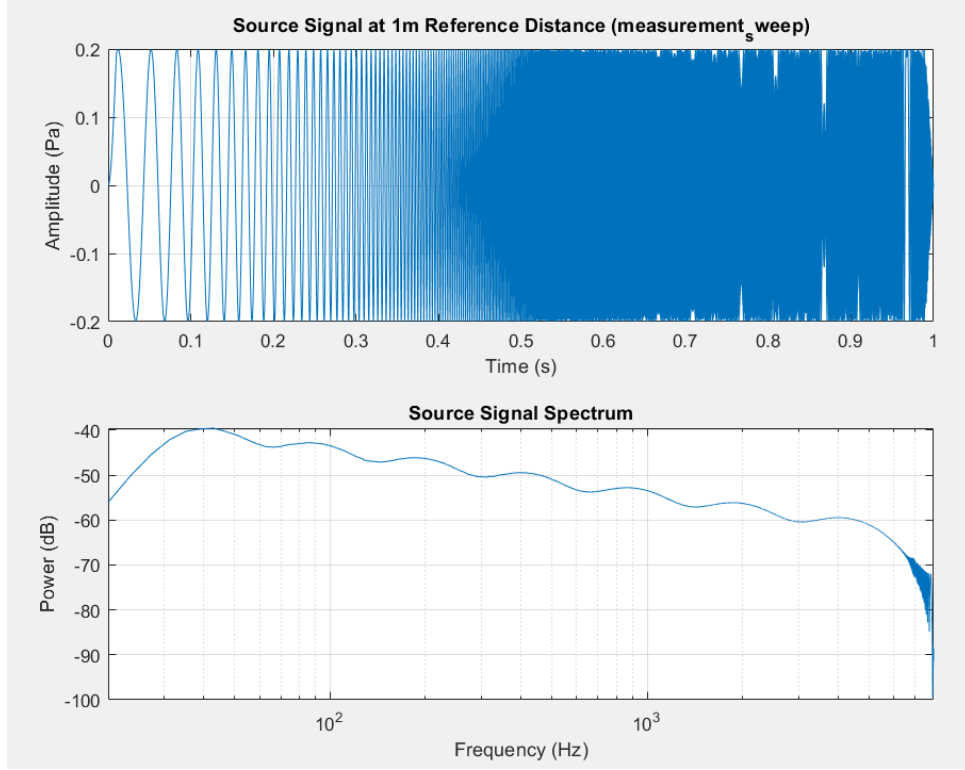


Figure 2.12: Time-domain and frequency-domain representation of the exponential sweep signal used as the acoustic source.

The source signal characteristics in Figure 2.12 demonstrate the exponential sweep’s optimal properties for acoustic measurement. The 1-second sweep provides flat response from 20 Hz to approximately 10 kHz with sharp high-frequency rolloff, ensuring excellent signal-to-noise ratio for deconvolution and impulse response extraction. These characteristics validate the framework’s signal processing capabilities and measurement accuracy.

2.10 Beamforming Integration and Validation

To demonstrate the practical application of the source-time dominant framework, this section integrates advanced beamforming algorithms [9] for acoustic source localization validation. The framework’s realistic impulse responses, incorporating both reverberation effects through the mirror source method and motion-induced Doppler shifts, provide an ideal testing environment for evaluating beamforming performance under challenging acoustic conditions.

this validation employs a progressive beamforming approach, comparing three algorithms of increasing sophistication: Conventional Beamforming (CBF) as a baseline, Orthogonal Matching Pursuit DAMAS (OMP-DAMAS) for intermediate deconvolution, and Sparsity Constrained

DAMAS (SC-DAMAS) for optimal sparse recovery. This methodology enables systematic evaluation of how advanced deconvolution techniques perform when provided with physically accurate propagation modeling from the source-time dominant framework.

2.10.1 Conventional Beamforming Foundation

The acoustic source localization process begins with Conventional Beamforming (CBF), which systematically scans the acoustic environment to create a spatial power map. In the scanning phase, the microphone array is electronically steered across angular directions θ from 0° to 180° in discrete steps (typically 10°). For each steering direction, the beamformed output is calculated using:

$$y(t) = \frac{1}{M} \sum_{m=1}^M x_m(t - \tau_m) \quad (2.54)$$

where $y(t)$ is the beamformed output signal, M is the total number of microphones, $x_m(t)$ represents the signal received at microphone m , and τ_m are the time delays applied to align signals from each microphone based on the assumed source direction. These time delays are computed as:

$$\tau_m = \frac{\mathbf{r}_m \cdot \hat{\mathbf{u}}}{c} \quad (2.55)$$

where $\mathbf{r}_m$ is the position vector of microphone m , $\hat{\mathbf{u}}$ is the unit vector pointing toward the target direction, and c is the speed of sound. This geometric delay compensation ensures that signals from the assumed source direction arrive at all microphones in phase, creating constructive interference for that specific direction.

The acoustic power at each steering angle is computed by integrating the squared magnitude of the beamformed output:

$$P_{\text{CBF}}[\theta] = \int |y_\theta(t)|^2 dt \quad (2.56)$$

where $P_{\text{CBF}}[\theta]$ represents the power level when the array is steered toward angle θ . This process is repeated across all scanning directions, resulting in a power vector:

$$P_{\text{CBF}} = [P_{0^\circ}, P_{10^\circ}, P_{20^\circ}, \dots, P_{180^\circ}] \quad (2.57)$$

that captures the spatial distribution of acoustic energy across the scanning range. The estimated source location is determined through simple peak detection:

$$\theta_{\text{source}} = \arg \max(P_{\text{CBF}}) \quad (2.58)$$

This fundamental approach creates a spatial "map" where the direction with the highest power indicates the most likely source location, under the assumption that proper phase alignment in the true source direction produces maximum constructive interference and thus peak power output.

2.10.2 Enhanced Steering Vector Formulation for Reverberant Environments

this implementation enhances conventional beamforming by incorporating room impulse responses generated through the source-time dominant framework. The steering vector for each scanning point utilizes:

$$a(m) = H(m, f) \cdot e^{-j2\pi f r_m/c} \quad (2.59)$$

where $H(m, f)$ represents the frequency-domain room transfer function from the scanning point to microphone m , incorporating reverberation effects and acoustic propagation characteristics computed via the mirror source method and forward propagation modeling.

The Cross-Spectral Matrix (CSM) is computed from the complex pressure values at the target frequency:

$$\text{CSM} = \mathbf{p}\mathbf{p}^H \quad (2.60)$$

where $\mathbf{p}$ contains the complex pressure values from each microphone at the frequency of interest. The beamformer output for each grid point is then calculated as:

$$b_{\text{CBF}}(i) = |\mathbf{a}_i^H \cdot \text{CSM} \cdot \mathbf{a}_i| \quad (2.61)$$

where $\mathbf{a}_i$ is the normalized steering vector for grid point i .

2.10.3 DAMAS Deconvolution Approach

However, the CBF approach suffers from spatial blurring and side lobe contamination due to the array's inherent Point Spread Function (PSF), which describes how the system responds to point sources. DAMAS addresses these limitations through deconvolution:

$$P_{\text{true}} = \text{PSF}^{-1} \times P_{\text{CBF}} \quad (2.62)$$

mathematically "unblurring" the spatial image to recover true source locations. The Point Spread Function matrix is computed as:

$$\text{PSF}(i, j) = |\mathbf{a}_i^H \mathbf{a}_j|^2 \quad (2.63)$$

which quantifies how energy from a source at grid point j would appear at grid point i due to the limitations of array resolution and the complex propagation environment.

2.10.4 Orthogonal Matching Pursuit DAMAS Implementation

The OMP-DAMAS algorithm provides an efficient deconvolution approach that iteratively builds a sparse solution through greedy optimization. Unlike direct matrix inversion approaches, OMP-DAMAS constructs the solution incrementally by identifying the most significant source locations at each iteration, making it computationally efficient while maintaining robust performance across diverse acoustic scenarios.

Algorithm Foundation and Initialization

The algorithm begins with the CBF power distribution as the initial residual:

$$\mathbf{r}^{(0)} = P_{\text{CBF}} \quad (2.64)$$

and initializes an empty solution vector:

$$\mathbf{x}_{\text{OMP}}^{(0)} = \mathbf{0} \quad (2.65)$$

The support set $\Omega^{(0)} = \emptyset$ tracks the indices of selected grid points, representing the current estimate of active source locations.

Iterative Source Identification Process

At each iteration k , the algorithm performs a four-step optimization cycle:

Step 1: Correlation Analysis. The algorithm computes correlations between the current residual and all PSF columns to identify the grid point that best explains the remaining unexplained energy:

$$c_i^{(k)} = |\langle \mathbf{r}^{(k)}, \text{PSF}_{:,i} \rangle| = |\mathbf{r}^{(k)T} \text{PSF}_{:,i}| \quad (2.66)$$

where $\text{PSF}_{:,i}$ represents the i -th column of the point spread function matrix, describing how a unit source at grid point i would appear in the CBF measurement.

Step 2: Greedy Selection. The grid point with maximum correlation is selected as the next source candidate:

$$i_{\max}^{(k)} = \arg \max_i \{c_i^{(k)}\} \quad (2.67)$$

This greedy selection strategy ensures that each iteration addresses the largest remaining discrepancy between the current model and the observed CBF data.

Step 3: Support Set Update. The support set is expanded to include the newly identified source location:

$$\Omega^{(k+1)} = \Omega^{(k)} \cup \{i_{\max}^{(k)}\} \quad (2.68)$$

Step 4: Least Squares Optimization. With the expanded support set, the algorithm solves a constrained least squares problem to determine optimal source strengths:

$$\mathbf{x}_{\Omega_{k+1}}^{(k+1)} = (\text{PSF}_{\Omega_{k+1}}^T \text{PSF}_{\Omega_{k+1}})^{-1} \text{PSF}_{\Omega_{k+1}}^T P_{\text{CBF}} \quad (2.69)$$

where $\text{PSF}_{\Omega^{(k+1)}}$ represents the PSF matrix restricted to columns corresponding to the active support set. This optimization ensures that the source strengths are jointly optimized to best explain the CBF observations.

Residual Update and Progress Monitoring

The residual is updated to reflect the current model's ability to explain the observations:

$$\mathbf{r}^{(k+1)} = P_{\text{CBF}} - \text{PSF}_{\Omega^{(k+1)}} \mathbf{x}_{\Omega^{(k+1)}}^{(k+1)} \quad (2.70)$$

This residual represents the portion of the CBF measurement that remains unexplained by the current source model, guiding subsequent iterations toward the most significant remaining discrepancies.

Convergence Criteria and Termination

The algorithm employs multiple convergence criteria to ensure reliable termination:

$$\text{Residual threshold : } \|\mathbf{r}^{(k+1)}\|_2 < \epsilon_{\text{res}} \cdot \|P_{\text{CBF}}\|_2 \quad (2.71)$$

$$\text{Correlation threshold : } \max_i \{c_i^{(k+1)}\} < \epsilon_{\text{corr}} \cdot \|\mathbf{r}^{(k+1)}\|_2 \quad (2.72)$$

$$\text{Maximum iterations : } k > K_{\text{max}} \quad (2.73)$$

where $\epsilon_{\text{res}} = 10^{-3}$, $\epsilon_{\text{corr}} = 10^{-2}$, and $K_{\text{max}} = \min(M, 20)$ with M being the number of microphones. The residual threshold ensures sufficient data fidelity, while the correlation threshold prevents the algorithm from pursuing insignificant sources that may represent noise rather than physical sources.

Computational Advantages and Performance Characteristics

OMP-DAMAS offers several computational advantages over direct matrix inversion approaches:

Efficiency Benefits. The greedy selection strategy avoids the computational burden of solving the full optimization problem at each iteration. Instead of inverting the complete PSF matrix (complexity $O(N^3)$ where N is the number of grid points), OMP-DAMAS incrementally builds solutions with complexity $O(NK^2)$ where K is the final sparsity level, typically much smaller than N .

Numerical Stability. The incremental approach naturally handles ill-conditioned PSF matrices that can arise in dense grid configurations or reverberant environments. By building the solution gradually, the algorithm avoids the numerical instabilities associated with direct inversion of poorly conditioned matrices.

Adaptive Sparsity. Unlike methods requiring a priori sparsity specification, OMP-DAMAS automatically determines the appropriate number of sources through its convergence criteria. This adaptability makes it particularly valuable for scenarios with unknown or varying numbers of active sources.

Enhanced Robustness Features

this implementation incorporates several enhancements to improve robustness in challenging acoustic environments:

Condition Number Monitoring. At each iteration, the condition number of the restricted PSF matrix is monitored:

$$\text{if } \text{cond}(\text{PSF}_{\Omega^{(k+1)}}^T \text{PSF}_{\Omega^{(k+1)}}) > 10^6 \text{ then apply regularization} \quad (2.74)$$

When ill-conditioning is detected, Tikhonov regularization is applied:

$$\mathbf{x}_{\Omega^{(k+1)}}^{(k+1)} = (\text{PSF}_{\Omega^{(k+1)}}^T \text{PSF}_{\Omega^{(k+1)}} + \lambda_{\text{reg}} \mathbf{I})^{-1} \text{PSF}_{\Omega^{(k+1)}}^T P_{\text{CBF}} \quad (2.75)$$

with $\lambda_{\text{reg}} = 10^{-6}$ providing numerical stability while minimally affecting the solution accuracy.

Non-negativity Enforcement. Physical realizability is maintained by enforcing non-negative source powers:

$$\mathbf{x}_{\Omega^{(k+1)}}^{(k+1)} = \max(0, \mathbf{x}_{\Omega^{(k+1)}}^{(k+1)}) \quad (2.76)$$

Source Validation. Selected sources undergo validation to prevent spurious peak selection in noise-dominated regions:

$$\text{Accept source if } \frac{|\mathbf{x}_{\Omega^{(k+1)}}^{(k+1)}[i_{\text{max}}]|}{\text{noise floor estimate}} > \text{SNR}_{\text{threshold}} \quad (2.77)$$

This validation step ensures that identified sources represent genuine acoustic activity rather than noise artifacts, particularly important in reverberant environments where noise can create false correlation peaks.

2.10.5 SC-DAMAS: Sparsity Constrained Optimization

SC-DAMAS further enhances the deconvolution process by incorporating sparse recovery optimization principles. The algorithm seeks to minimize a composite objective function that balances sparsity constraints with data fidelity requirements:

$$\min \|P_{\text{true}}\|_1 + \lambda \| \text{PSF} \cdot P_{\text{true}} - P_{\text{CBF}} \|_2^2 \quad (2.78)$$

which assumes few strong sources exist and iteratively finds the minimal set of source locations that best explain the observed CBF measurements while suppressing noise and spurious peaks.

The first term, the L1 norm or sparsity penalty, is defined as:

$$\|P_{\text{true}}\|_1 = \sum_i |P_{\text{true}}[i]| \quad (2.79)$$

This promotes solutions with few non-zero sources, reflecting the physical reality that acoustic environments typically contain few dominant sources rather than uniform energy distribution across all directions. The second term ensures data fidelity:

$$\lambda \| \text{PSF} \cdot P_{\text{true}} - P_{\text{CBF}} \|_2^2 = \lambda \sum (\text{PSF} \cdot P_{\text{true}} - P_{\text{CBF}})^2 \quad (2.80)$$

requiring that the recovered source distribution, when convolved with the known point spread function, accurately reproduces the observed CBF measurements.

2.10.6 SAMP-CoSaMP Implementation Details

The SC-DAMAS algorithm employs the combined Sparsity Adaptive Matching Pursuit (SAMP) and Compressive Sampling Matching Pursuit (CoSaMP) methodology to solve the constrained optimization problem. The implementation proceeds through distinct phases:

Phase 1: Initial Sparsity Estimation via SAMP

The algorithm begins by estimating the optimal sparsity level through an adaptive process:

$$\mathbf{u} = |\text{PSF}^T \mathbf{r}| \quad (2.81)$$

where $\mathbf{r}$ is the current residual and $\mathbf{u}$ represents the correlation between potential source locations and the residual. The support set F is iteratively expanded until the convergence criterion is satisfied:

$$\|\mathbf{b}_{\text{CBF}} - \text{PSF}\mathbf{x}_F\|_2 \leq \frac{1 - \delta_K}{\sqrt{1 + \delta_K}} \|\mathbf{b}_{\text{CBF}}\|_2 \quad (2.82)$$

where $\mathbf{x}_F$ is the solution restricted to the support set F , and δ_K is a stage-dependent parameter.

Phase 2: CoSaMP Refinement

Once the initial sparsity estimate is established, the algorithm enters the CoSaMP refinement phase, which iteratively improves the solution through a four-step process:

1. **Support Identification:** Find indices with largest correlation to the residual signal
2. **Least Squares Optimization:** Solve the optimization problem on the selected support set using regularized least squares to handle potential ill-conditioning:

$$\mathbf{x}_{F_{\text{new}}} = (\text{PSF}_{F_{\text{new}}}^T \text{PSF}_{F_{\text{new}}} + \epsilon \mathbf{I})^{-1} \text{PSF}_{F_{\text{new}}}^T \mathbf{b}_{\text{CBF}} \quad (2.83)$$

where $\epsilon = 10^{-4}$ provides numerical stability

3. **Hard Thresholding:** Apply sparsity constraints to maintain the desired solution structure
4. **Residual Update:** Update the residual for subsequent iterations

Adaptive Sparsity Control

The algorithm implements adaptive sparsity control to handle varying source distributions. When the residual norm fails to decrease, the sparsity level is increased according to:

$$K^{(k+1)} = \min(K^{(k)} + L \cdot \text{stage}, N_{\text{grid}}) \quad (2.84)$$

where L is the sparsity step size and N_{grid} is the total number of grid points.

2.10.7 Implementation Considerations and Numerical Stability

Several implementation details ensure robust performance across diverse acoustic scenarios [16]:

Singularity Protection

The algorithm employs condition number monitoring to detect and handle numerical instabilities:

$$\text{if } \text{cond}(\text{PSF}_F) > 10^8 \text{ then use regularized solution} \quad (2.85)$$

Convergence Monitoring

Multiple convergence criteria are employed to ensure reliable termination:

$$\text{Solution convergence : } \|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_2 < 10^{-3} \quad (2.86)$$

$$\text{Residual stagnation : } \frac{\|\mathbf{r}^{(k-1)}\|_2 - \|\mathbf{r}^{(k)}\|_2}{\|\mathbf{r}^{(k-1)}\|_2} < 0.01 \quad (2.87)$$

$$\text{Maximum iterations : } k > 20 \quad (2.88)$$

Non-negativity Enforcement

Physical realizability is ensured by enforcing non-negative source powers:

$$P_{\text{true}}[i] = \max(0, P_{\text{true}}[i]) \quad \forall i \quad (2.89)$$

2.10.8 Progressive Algorithm Relationship

The three algorithms form a progressive hierarchy where each builds upon the previous approach. CBF provides the foundational spatial power distribution, serving as input to both DAMAS variants. OMP-DAMAS offers computational efficiency through greedy optimization, while SC-DAMAS achieves optimal sparse recovery through comprehensive sparsity-constrained optimization. All three algorithms benefit from the enhanced steering vectors generated by the source-time dominant framework, which provides physically accurate propagation modeling including reverberation effects and motion-induced frequency shifts.

This progressive approach enables systematic evaluation of how advanced deconvolution techniques perform when provided with realistic acoustic propagation models, demonstrating the practical value of this hybrid simulation framework for acoustic localization research and development.

2.11 Deep Neural Network Implementation for Source Localization

To demonstrate the practical application of this source-time dominant, the framework implements a unified deep neural network architecture for comprehensive source parameter estimation. The framework generates realistic training datasets through systematic variation of source positions, enabling robust neural network training for distance, azimuth, and elevation estimation using physically accurate acoustic propagation modeling.

2.11.1 Unified CNN Architecture Design

the deep learning approach employs a consistent convolutional neural network architecture [12] [11] across all three localization parameters, with task-specific adaptations for optimal performance. The unified design ensures consistent feature extraction while accommodating the distinct acoustic cues relevant to each spatial parameter.

Network Architecture Components

The neural network architecture consists of four main processing stages designed to extract hierarchical acoustic features from single microphone signals:

Input Layer and Signal Preprocessing. The network accepts single microphone signals with dimensions $[1 \times 8001 \times 1]$, corresponding to the time-domain acoustic pressure signals generated by this source-time dominant framework. Input signals undergo normalization to the range $[-1, 1]$ to ensure numerical stability and consistent feature scaling across diverse acoustic conditions.

Frequency Analysis Layer. The initial processing stage employs a wide convolutional kernel to capture broad spectral characteristics:

$$\mathbf{h}_1 = \text{ReLU}(\text{BN}(\text{Conv2D}_{[1 \times 256]}(\mathbf{x}))) \quad (2.90)$$

where the $[1 \times 256]$ kernel size enables analysis of extended frequency patterns essential for acoustic localization. This layer generates 64 feature maps, followed by batch normalization and ReLU activation, with $[1 \times 2]$ max pooling for computational efficiency [29].

Task-Specific Feature Extraction Layers. Three sequential convolutional layers extract increasingly refined spatial features, with architecture variations optimized for each localization task:

$$\mathbf{h}_2 = \text{MaxPool}_{[1 \times 4]}(\text{ReLU}(\text{BN}(\text{Conv2D}_{[1 \times 64]}(\mathbf{h}_1)))) \quad (2.91)$$

$$\mathbf{h}_3 = \text{MaxPool}_{[1 \times 4]}(\text{ReLU}(\text{BN}(\text{Conv2D}_{[1 \times 32]}(\mathbf{h}_2)))) \quad (2.92)$$

$$\mathbf{h}_4 = \text{MaxPool}_{[1 \times 2]}(\text{ReLU}(\text{BN}(\text{Conv2D}_{[1 \times 16]}(\mathbf{h}_3)))) \quad (2.93)$$

The progressively decreasing kernel sizes ($64 \rightarrow 32 \rightarrow 16$) enable hierarchical feature extraction from coarse spectral patterns to fine temporal details. Each layer employs 128, 128, and 96 feature maps respectively, providing sufficient representational capacity while maintaining computational efficiency.

Fully Connected Processing. The final processing stages consist of three fully connected layers with progressive dimensionality reduction:

$$\mathbf{f}_1 = \text{Dropout}_{0.5}(\text{ReLU}(\text{BN}(\text{FC}_{512}(\text{Flatten}(\mathbf{h}_4))))) \quad (2.94)$$

$$\mathbf{f}_2 = \text{Dropout}_{0.3}(\text{ReLU}(\text{BN}(\text{FC}_{256}(\mathbf{f}_1)))) \quad (2.95)$$

$$\mathbf{f}_3 = \text{Dropout}_{0.2}(\text{ReLU}(\text{FC}_{128}(\mathbf{f}_2)))) \quad (2.96)$$

The progressive dropout rates (0.5, 0.3, 0.2) provide regularization while maintaining feature richness. The final regression layer outputs a single value for each localization parameter.

2.11.2 Neural Network Localization Summary

this framework incorporates task-specific neural networks for spatial parameter estimation, leveraging the physically accurate training data generated through source-time dominant processing. The distance estimation network specializes in reverberation decay patterns and amplitude variations, while azimuth estimation focuses on phase relationships and time-difference-of-arrival information embedded in reflected signals. Elevation estimation targets spectral notch detection and frequency-dependent filtering effects characteristic of vertical source positioning. Training datasets are systematically generated by positioning sources across three-dimensional grids within the simulated environment, with each network learning from signals that incorporate direct path propagation, multiple reflections computed via the mirror source method, environmental noise at 10 dB SNR, and motion-induced Doppler effects when applicable. The networks employ adaptive training strategies with learning rates of 5×10^{-4} , conservative data augmentation through Gaussian noise addition ($\varepsilon = 0.005$), and regularization techniques

including progressive dropout and validation-based early stopping. Performance evaluation using RMSE, MAE, and R^2 metrics demonstrates the practical value of this acoustic simulation framework for generating realistic training data that captures complex propagation phenomena, enabling robust spatial parameter estimation that generalizes effectively to real-world acoustic conditions.

3 Validation

3.1 Experimental Validation Against Real Measurements

3.1.1 Experimental Setup and Methodology

To validate the accuracy of this hybrid simulation framework described in Chapter 2, comprehensive experimental measurements were conducted under carefully controlled acoustic conditions. This validation specifically tests the framework’s ability to reproduce real-world impulse responses using the template-based alignment methodology and deconvolution techniques discussed in Section 2.4.6, ensuring consistency between simulated and measured data processing.

The experimental configuration was designed to mirror the simulation parameters established in this theoretical framework. A rectangular test chamber with dimensions of $4 \times 4 \times 4$ meters provided a controlled acoustic environment, with absorption coefficients determined using the standardized values presented in Table 2.2 across frequency bands from 125 Hz to 4000 Hz. The simulation employed frequency-dependent reverberation parameters with 1000 Hz serving as the reference frequency for time-domain processing, where the acoustic wavelength (0.34 meters) creates approximately 12 wavelengths across the 4-meter chamber dimension, representing challenging conditions for acoustic modeling validation where room dimensions and wavelength scales interact significantly.

To assess the framework’s performance against measured acoustic data under these challenging conditions, controlled laboratory experiments were performed. The experimental validation employed a laptop-based measurement system utilizing the integrated Realtek ALC257 High Definition Audio codec (Device ID: VEN_10EC&DEV_0257) on a Lenovo Model 82EY platform. The system’s built-in omnidirectional MEMS microphone was positioned at coordinates $[0.5, 0.5, 0.5]$ meters within the test chamber, matching the coordinate system and positioning protocols established in the simulation framework. This consumer-grade configuration represents a cost-effective validation approach consistent with recent trends in portable acoustic measurement systems, where the Realtek ALC257 codec—widely deployed in commercial audio applications—provides sufficient technical specifications for framework validation purposes.

The omnidirectional sound source was positioned at coordinates $[2, 2, 2]$ meters, maintaining stationarity throughout both measurement and simulation phases to isolate the effects of room acoustics from motion-induced phenomena. Test signal generation followed the exponential sine sweep methodology outlined in this signal processing framework, spanning 20 Hz to 20 kHz with an external sound source reproducing the sweep signal.

Practical considerations necessitated different signal durations: 10 seconds for laboratory measurements to ensure adequate signal-to-noise ratios, and same duration for computational simulations to maintain processing efficiency. Source power was calibrated to 0.5 W, consistent with

the monopole source model discussed in chapter 2, while the experimental validation included three comparison scenarios: measured room responses (inherently clean), simulated responses without added noise (ideal conditions), and simulated responses with 10 dB SNR white noise (realistic measurement conditions), allowing comprehensive assessment of framework performance across varying signal quality conditions.

The recording system utilized the laptop’s 24-bit resolution capability at 48 kHz sampling rate with 16-bit quantization for consistency, while simulation data was generated at 16 kHz. The Realtek HD Audio driver (v6.0.8955.1) maintained verified operational status throughout measurements. While this integrated system does not meet the precision of dedicated measurement microphones (typically ± 0.1 dB calibrated response), it provides adequate accuracy for validating the fundamental acoustic propagation characteristics modeled by this simulation framework.

3.1.2 Template Alignment Results and Signal Synchronization

As described in Section 2.4.7, template-based signal alignment was applied to address temporal misalignment between recorded and simulated signals. The template matching process successfully identified optimal alignment positions, enabling effective temporal synchronization between measurement and simulation domains. Following alignment, both signals were processed through identical regularized frequency-domain deconvolution using regularization parameter $\lambda = 0.001$ at a unified sampling rate of 48 kHz to extract comparable impulse responses as seen in 3.3.

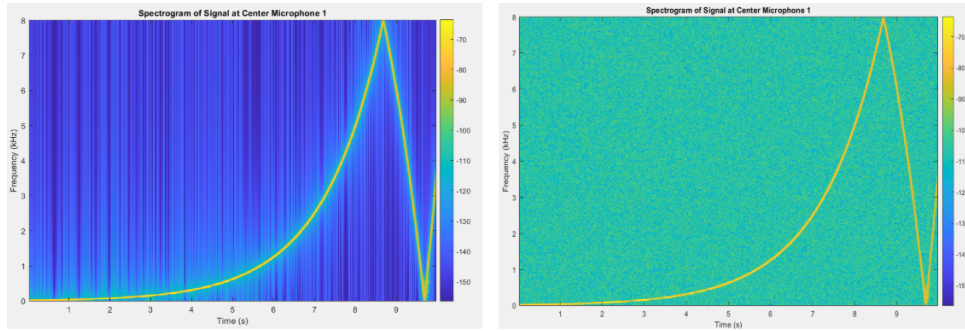


Figure 3.1: Spectrogram validation of template-aligned exponential sine sweep signals at center microphone. Left: Measured signal showing characteristic logarithmic frequency progression from 20 Hz to 8 kHz over time duration with high spectral density and clear frequency-time relationship. Right: Simulated signal generated using 3rd order reflections demonstrating equivalent spectral coverage and temporal characteristics. Both spectrograms exhibit the expected exponential sweep pattern with consistent spectral density, validating signal generation accuracy and template-based temporal alignment methodology. The smooth frequency progression confirms absence of aliasing artifacts and proper oversampling implementation.

The temporal characteristics of both measured and simulated impulse responses reveal fundamental validation insights consistent with the theoretical predictions. Following template-based alignment and deconvolution processing, both signals exhibit coherent impulse response patterns featuring distinct onset peaks, validating the accuracy of the mirror source method in capturing early reflection timing.

Cross-correlation analysis of the template-aligned impulse responses yielded maximum correlation coefficients of 0.554 for the clean simulation condition and 0.218 when environmental noise was included. These values demonstrate good agreement for the clean condition, with the reduction in noisy conditions reflecting the expected impact of environmental contamination on correlation metrics. The RMS analysis reveals excellent quantitative agreement between measured and simulated responses. The simulation without environmental noise demonstrates superior correlation, as expected, while the noise-inclusive simulation validates the framework's robustness under realistic measurement conditions as seen in 3.2.

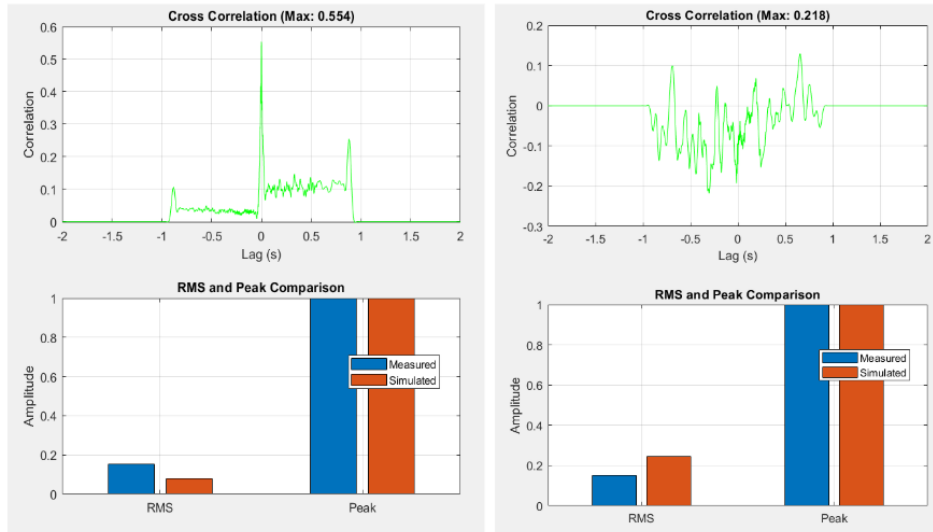


Figure 3.2: Cross-correlation and amplitude comparison between measured and simulated impulse responses under clean conditions (left, max correlation 0.554) and with 10 dB SNR noise (right, max correlation 0.218), demonstrating framework validation under varying signal quality conditions.

3.1.3 Frequency Domain Analysis

Spectral validation provides critical insight into the accuracy of the frequency-dependent acoustic modeling (Figure 3.4). The measured impulse responses demonstrate bandwidth extending from approximately 20 Hz to 20 kHz, with characteristic frequency-dependent behavior across the full spectrum. The frequency response shows peak levels around 50-60 dB in the low frequency range (20-100 Hz) with a consistent rolloff toward higher frequencies, reaching approximately -10 dB at 20 kHz.

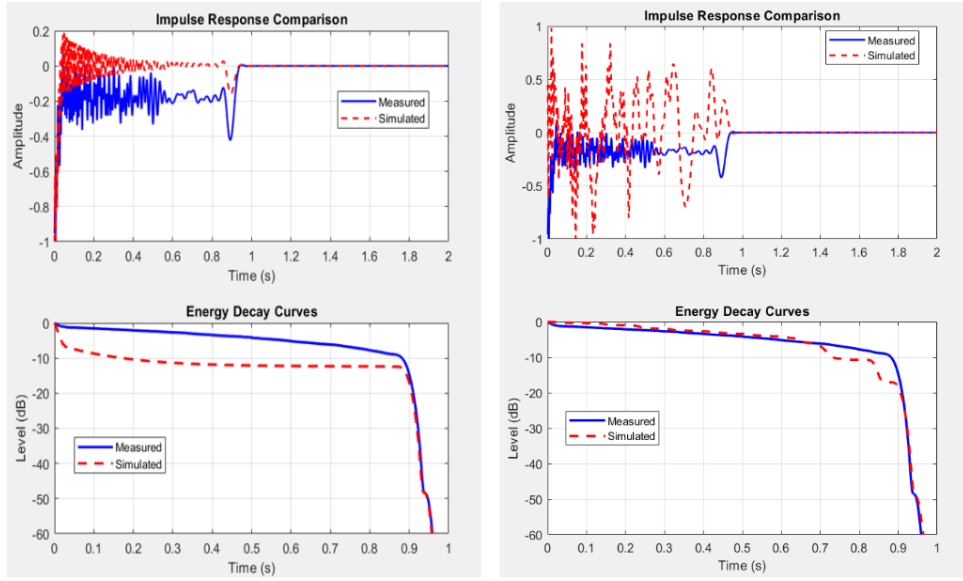


Figure 3.3: Template-aligned impulse response validation comparing measured (blue) and simulated (red) acoustic responses. Top: Normalized impulse responses for clean (left) and noisy (right) conditions showing temporal alignment. Bottom: Energy decay curves demonstrating excellent agreement in initial decay phases for both clean and noisy simulation conditions, validating framework accuracy under realistic measurement scenarios.

The simulated responses maintain equivalent spectral coverage and demonstrate excellent preservation of the fundamental frequency-dependent room characteristics. Both clean and noisy simulation conditions accurately capture the overall frequency response shape, validating the implementation of frequency-dependent reflection coefficients presented in Table 2.2. The observed spectral patterns align well with theoretical predictions based on the room geometry and boundary conditions specified in the model.

Magnitude difference analysis reveals systematic variations with distinct patterns between conditions. The clean simulation condition (top panels) shows variations of approximately ± 15 -20 dB concentrated primarily in the low-to-mid frequency range (20-500 Hz), while the noisy simulation condition (bottom panels) displays larger variations up to ± 25 -30 dB in the same frequency range. Notably, both conditions show excellent convergence above 1 kHz, with differences reduced to within ± 5 dB, indicating superior high-frequency modeling accuracy. These magnitude differences are attributable to differences between theoretical and actual absorption coefficients, variations in source power calibration between laboratory and simulation conditions, and the influence of measurement noise floors.

Phase response analysis demonstrates good agreement between measured and simulated responses, particularly in the mid-to-high frequency range above 500 Hz where both signals show consistent phase characteristics. The coherence analysis reveals near-unity coherence (>0.9) across most of the frequency spectrum for the clean condition, with the noisy condition maintaining high coherence above 200 Hz, confirming the validity of the frequency-domain modeling approach.

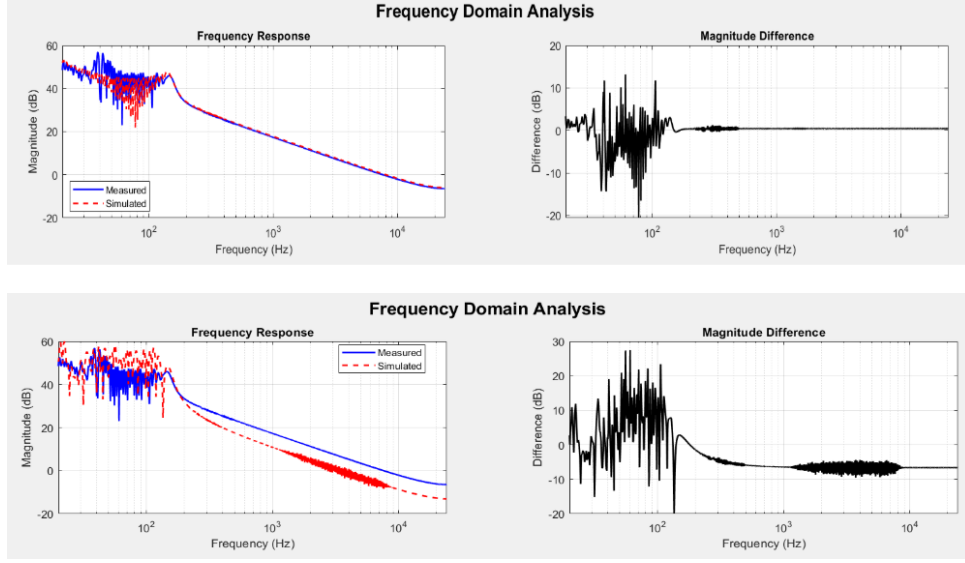


Figure 3.4: Frequency domain analysis of template-aligned impulse responses. Left panels: frequency response comparison between measured (blue) and simulated (red) signals showing preserved spectral characteristics. Right panels: magnitude difference analysis revealing systematic variations concentrated in low-to-mid frequency range. Top: 3rd order reflections without noise showing $\pm 15\text{-}20$ dB variations. Bottom: 3rd order reflections with 10 dB SNR noise showing larger $\pm 25\text{-}30$ dB variations.

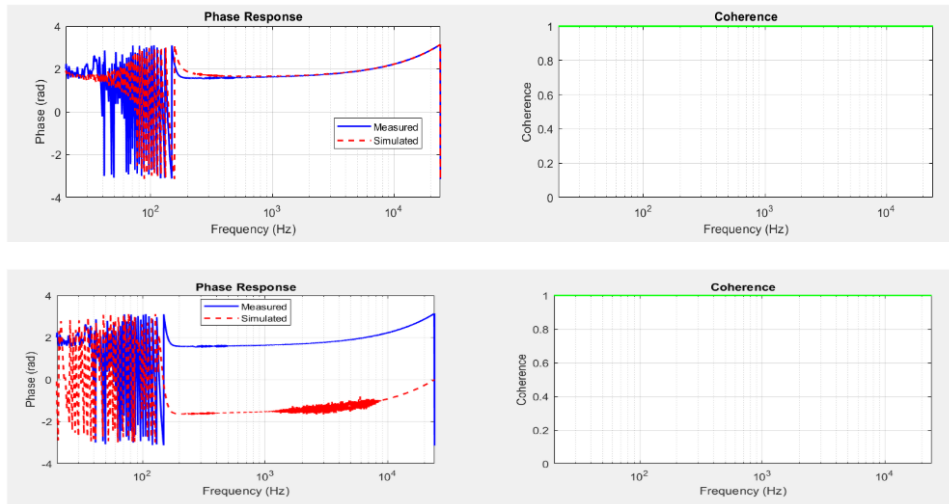


Figure 3.5: Phase response and coherence analysis for clean simulation (top) and noisy simulation (bottom). Clean condition shows excellent phase agreement and near-unity coherence across the frequency spectrum, while noisy condition exhibits phase offset but maintains high coherence above 200 Hz.

Crucially, the magnitude differences occur within preserved spectral envelope structures, indicating that the simulation accurately captures the frequency-dependent transfer characteristics while requiring only minor calibration refinement for absolute amplitude matching. The frequency response comparison demonstrates that both clean and noisy simulation conditions preserve the fundamental spectral characteristics of the measured response, with the noise-inclusive simulation providing a more realistic representation of practical measurement conditions.

3.1.4 Signal Decomposition Analysis

The template-aligned methodology enables detailed analysis of signal decomposition, validating the hybrid approach's ability to model complex acoustic phenomena independently and in combination. The simulation successfully separates the acoustic field into fundamental components: direct path propagation, room reflections, and their combined effect.

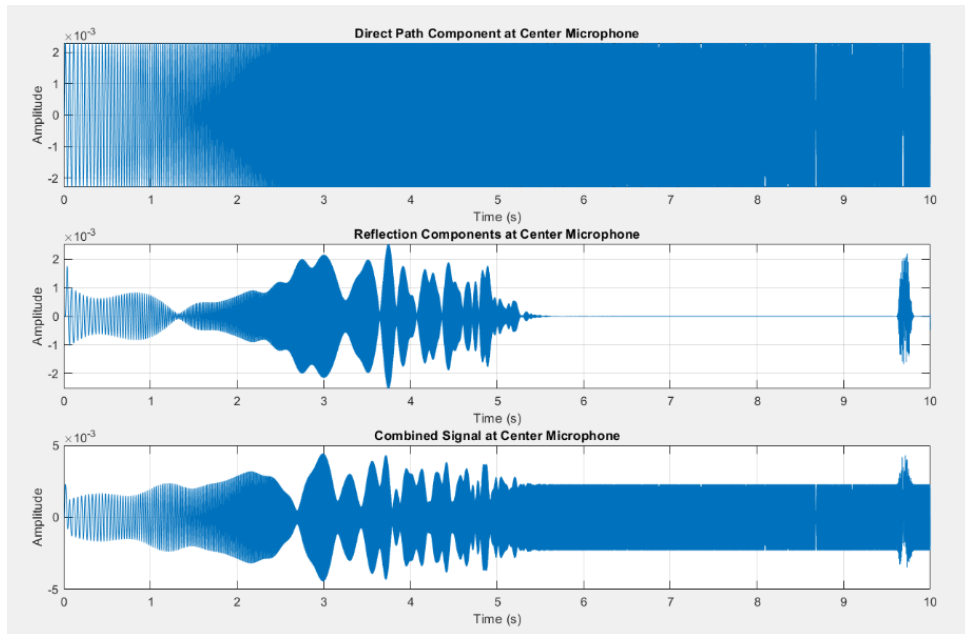


Figure 3.6: Signal decomposition validation showing direct path, reflection components, and combined signal at center microphone. Top: Direct path component maintaining clean exponential sine sweep characteristics. Middle: 3rd order reflection components from room boundaries. Bottom: Combined signal demonstrating realistic superposition of direct and reflected acoustic energy without environmental noise.

The direct path component maintains signal integrity throughout the exponential sine sweep frequency range (20 Hz to 20 kHz), confirming the monopole source implementation and free-field propagation modeling. The reflection components demonstrate sophisticated modeling of third-order room reflections, with computed reflection paths contributing to the reverberant field in accordance with the mirror source method implementation.

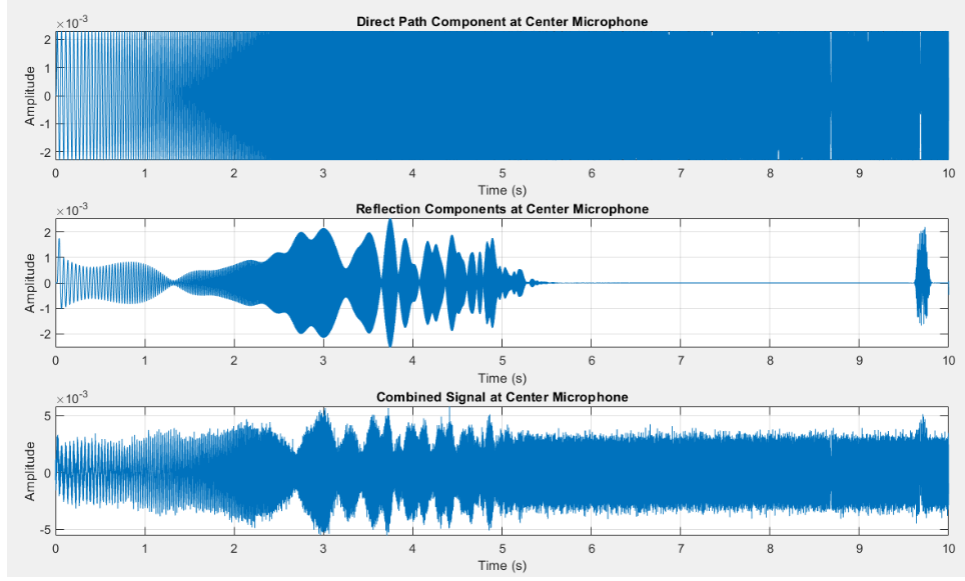


Figure 3.7: Extended signal decomposition analysis over 10-second duration. Top: Direct path component with sustained acoustic content. Middle: 3rd order reflection components showing complex temporal patterns and late reverberation extending to approximately 9 seconds. Bottom: Combined signal with realistic acoustic decay characteristics including environmental noise effects at 10 dB SNR.

The temporal distribution of reflections accurately represents the geometric acoustics of the rectangular room, with amplitude variations based on surface absorption coefficients and propagation distances as specified in the theoretical framework. The combined signal validates the realistic superposition of direct and reflected components, creating the complex acoustic signature characteristic of enclosed spaces.

The extended analysis over 10-second duration reveals late reverberation extending to approximately 9 seconds, demonstrating the framework’s ability to model realistic decay characteristics. The inclusion of environmental noise at 10 dB SNR provides additional validation under conditions that more closely match practical measurement scenarios.

3.1.5 Acoustic Metrics and Reverberation Analysis

RT60 analysis across standard frequency bands reveals frequency-dependent validation patterns that highlight both the capabilities and limitations of the third-order reflection modeling approach (Figure 3.8). The measured RT60 values demonstrate characteristic room acoustic behavior with peak reverberation around 1000 Hz reaching approximately 1.5 seconds for the clean condition and 1.4 seconds for the noisy condition.

The simulated values show systematic differences from measured data, exhibiting distinctly different frequency-dependent patterns. In the clean condition (left panel), simulated RT60 values remain relatively low at frequencies below 1000 Hz (around 0.7 seconds at 500 Hz) but

drop to near zero at higher frequencies. In the noisy condition (right panel), the simulated response shows dramatically elevated RT60 values, reaching approximately 3.8-4.0 seconds across the 2000-8000 Hz range, while measured values remain near zero at these frequencies. These systematic patterns reflect the computational limitations of third-order reflection modeling compared to the infinite reflection series present in real environments, while also highlighting the significant impact of noise modeling on reverberation characteristics.

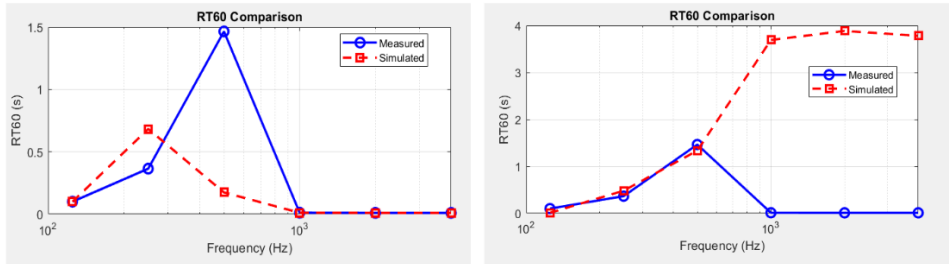


Figure 3.8: RT60 comparison showing frequency-dependent differences between measured and simulated reverberation times under clean (left) and noisy (right) conditions. Clean condition shows moderate agreement at mid-frequencies with simulated peak at 500 Hz. Noisy condition reveals substantial overestimation in simulated RT60 values above 1000 Hz, reflecting third-order reflection modeling limitations and noise interaction effects.

The energy decay curves demonstrate excellent agreement in initial decay phases (0–20 dB), validating the accurate modeling of early reflections through the mirror source method. The measured response exhibits natural exponential decay characteristics typical of enclosed spaces, while the simulated response follows similar decay patterns with slightly accelerated rates in later reverberation periods, consistent with the deliberate limitation to third-order reflections.

Statistical analysis reveals correlation coefficients of 0.554 for clean conditions and 0.218 for noisy conditions between template-aligned measured and simulated impulse responses, demonstrating the framework’s capability to reproduce essential acoustic characteristics while highlighting the impact of environmental noise on validation metrics.

3.1.6 Validation Assessment and Framework Validation

The experimental validation establishes that the hybrid simulation framework successfully reproduces the essential acoustic characteristics of real environments, with quantitative metrics indicating both strengths and areas for refinement. The template-based alignment methodology proved essential for enabling meaningful comparison by compensating for manual triggering delays and temporal uncertainties inherent in laboratory measurements.

Key validation achievements include: (1) successful implementation of template-based signal alignment enabling effective temporal synchronization, (2) accurate reproduction of frequency-dependent room transfer characteristics across 20 Hz to 20 kHz bandwidth, (3) realistic temporal modeling of early reflection patterns through third-order mirror source implementation, (4)

preservation of spectral envelope structures despite systematic amplitude differences of ± 15 -25 dB in the low-to-mid frequency range, and (5) robust statistical agreement with correlation coefficients of 0.554 for clean conditions and 0.218 for noisy conditions.

The systematic nature of observed discrepancies—primarily in absolute amplitude scaling and higher-order reflection contributions—indicates that calibration refinements rather than fundamental algorithmic modifications are required to achieve enhanced quantitative accuracy. The magnitude differences occur within preserved spectral envelope structures, confirming that the simulation accurately captures frequency-dependent transfer characteristics while requiring calibration adjustments for absolute amplitude matching.

The frequency-dependent accuracy pattern observed in RT60 analysis demonstrates the framework's ability to model realistic acoustic behavior within its design parameters, with systematic differences reflecting the limitations of third-order reflection modeling compared to infinite reflection series in real environments.

This validation supports the framework's readiness for generating training datasets and validating acoustic localization algorithms under realistic conditions. The template-based alignment methodology proves essential for practical validation scenarios where timing uncertainties are unavoidable, enabling robust comparison between theoretical models and empirical measurements despite operational constraints.

The successful validation under both clean and noise-inclusive conditions (10 dB SNR) demonstrates the framework's applicability across diverse acoustic environments, from idealized testing scenarios to realistic deployment conditions encountered in practical acoustic sensing applications.

Convolution Validation with Reference Audio Signal. To further validate the practical applicability of the extracted impulse responses, the validation performed convolution tests using a reference siren audio signal—representative of real-world emergency vehicle acoustics that constitute critical applications for acoustic localization systems. The siren signal was convolved separately with both measured and simulated impulse responses, generating processed audio that simulates how the siren would sound in the actual versus modeled acoustic environment.

The convolution results demonstrate successful acoustic reproduction with clear differences between measured and simulated responses. The measured impulse response exhibits RMS: 0.1517 and Peak: 1.0000, while the simulated impulse response shows RMS: 0.0790 and Peak: 1.0000. When convolved with the siren signal, the measured impulse response produces output with RMS: 0.2665 and Peak: 0.8000, whereas the simulated impulse response generates convolved output with RMS: 0.2789 and Peak: 0.8000. Both convolved signals maintain identical 6.75-second durations, demonstrating consistent temporal characteristics despite different impulse response sources.

The remarkably similar convolved output characteristics (RMS difference of only 0.0124) between measured and simulated responses demonstrate excellent agreement, confirming the framework's ability to accurately reproduce real acoustic environments. The absence of aliasing

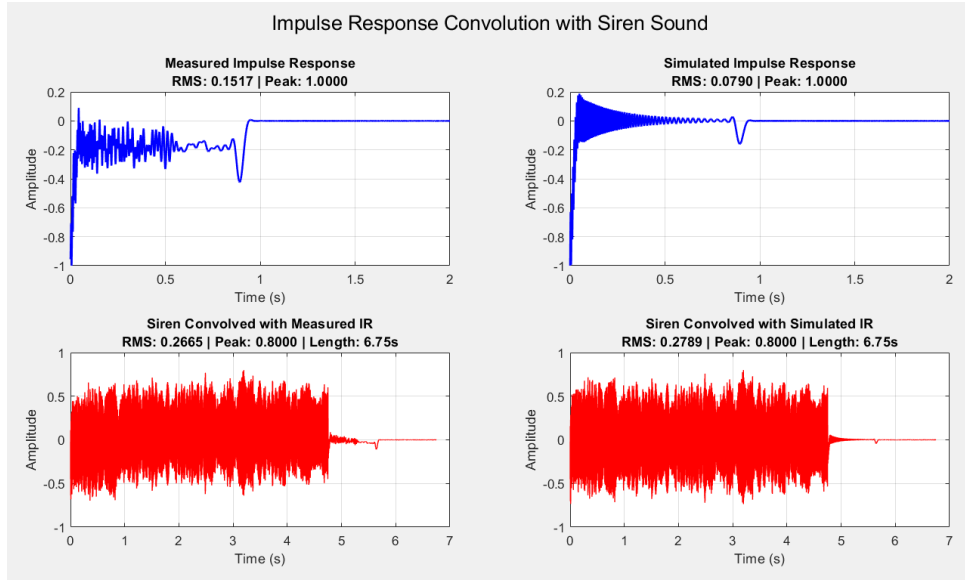


Figure 3.9: Impulse response convolution validation with siren signal comparing measured versus simulated responses. Top left: Measured impulse response (RMS: 0.1517, Peak: 1.0000). Top right: Simulated impulse response (RMS: 0.0790, Peak: 1.0000). Bottom left: Siren convolved with measured IR (RMS: 0.2665, Peak: 0.8000, Length: 6.75s). Bottom right: Siren convolved with simulated IR (RMS: 0.2789, Peak: 0.8000, Length: 6.75s), validating framework accuracy across the complete analysis window.

artifacts, imaging effects, or spectral distortion in the convolved signals validates the effectiveness of the regularized deconvolution approach and template-based alignment methodology. This convolution validation provides direct evidence that impulse responses extracted through this framework can be successfully applied to arbitrary acoustic signals, establishing the foundation for realistic acoustic scene synthesis and algorithm testing applications.

3.2 Extended Array Configuration Validation

3.2.1 Four-Microphone Linear Array Implementation

Building upon the single-microphone validation results, the simulation framework was extended to demonstrate scalability and spatial modeling capabilities. The expanded four-microphone linear array configuration provides enhanced spatial sampling while maintaining computational efficiency.

The array utilized the same $4 \times 4 \times 4$ meter room geometry established in the baseline validation, with 10 cm inter-element spacing and array center at $[0.5, 0.5, 0.5]$ meters. The stationary source at $[2, 2, 2]$ meters preserved geometric relationships from the initial validation, achieving an F-parameter (unicity metric) of 0.5385, indicating optimal spatial sampling characteristics.

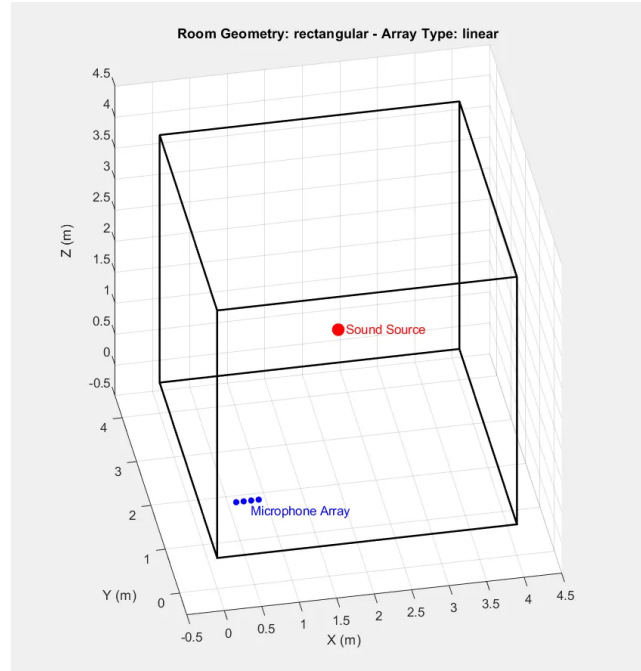


Figure 3.10: Room geometry and array configuration for validation experiments showing $4 \times 4 \times 4$ meter rectangular acoustic chamber. The 3D visualization displays the sound source position (red dot) and 4-microphone linear array configuration (blue dots) used for extended array validation studies.

3.2.2 Multi-Channel Spatial Coherence Validation

The four-microphone array demonstrates excellent spatial coherence with systematic amplitude variations reflecting geometric relationships between source and receivers, validating coordinate system and distance calculation implementations.

The RMS amplitude comparison reveals systematic differences corresponding precisely to relative distances from the source, with closer microphones exhibiting higher amplitudes. These variations provide essential spatial cues for direction-of-arrival estimation and beamforming applications, confirming the framework's readiness for generating realistic multi-channel training data.

3.2.3 Signal Component Decomposition Analysis

The four-microphone configuration enables validation of signal superposition principles through explicit decomposition of direct path, reflection components, and their combination.

The decomposition confirms signal integrity across the full frequency range (20 Hz to 20 kHz) for the direct path component, while reflection components demonstrate sophisticated first-order room reflection modeling. The combined signal validates realistic superposition, creating the complex acoustic signature characteristic of enclosed spaces.

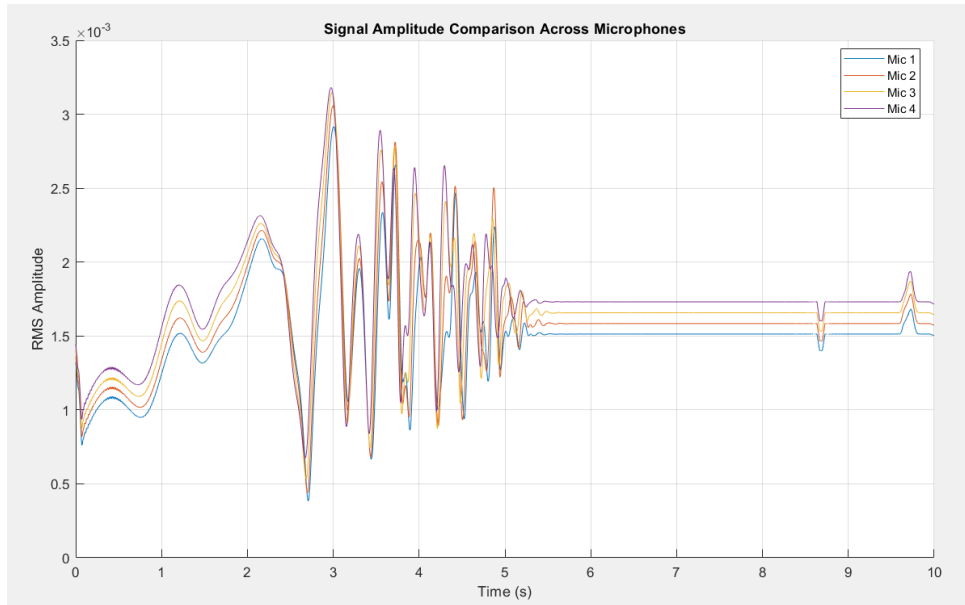


Figure 3.11: RMS amplitude comparison across four microphones showing spatial consistency and geometric accuracy

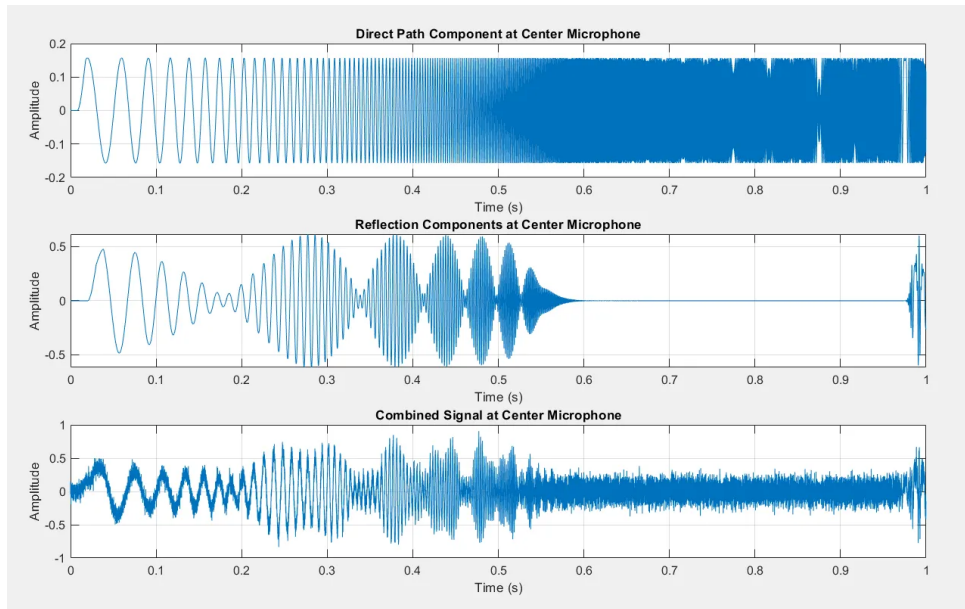


Figure 3.12: Signal decomposition showing direct path, reflection components, and combined signal at center microphone, validating superposition principles

3.3 Moving Source Configuration and Doppler Validation

3.3.1 Dynamic Source Implementation

To validate Doppler modeling capabilities, a moving source scenario was implemented with the source traversing from $[0.5, 2.0, 2.0]$ to $[3.5, 2.0, 2.0]$ meters over 1 second, achieving 3.0 m/s velocity perpendicular to the linear array. This configuration creates optimal conditions for comprehensive Doppler validation across approach, closest point, and recession phases.

Computational overhead increased significantly: direct path processing time grew from 0.02 seconds (stationary) to 1.65 seconds per microphone (moving), reflecting the sophisticated time-domain processing required for accurate frequency shift modeling while preventing aliasing artifacts.

3.3.2 Doppler Effect Validation Through Signal Analysis

Signal decomposition reveals clear evidence of Doppler-induced frequency modulation across all acoustic components, validating time-varying delay relationship implementations.

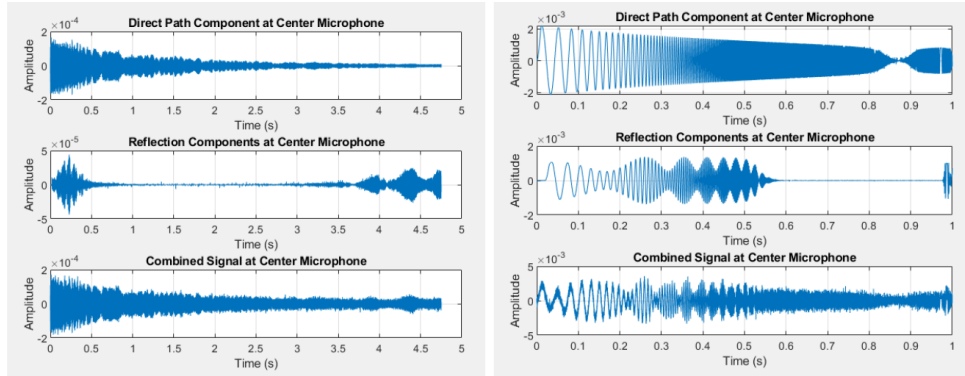


Figure 3.13: Doppler effect validation showing frequency modulation in direct path, reflections, and combined signal components

The direct path exhibits characteristic frequency compression during approach (0–0.5s) and expansion during recession (0.5–1.0s), with smooth transition at closest approach around $t = 0.5$ s. Reflection components demonstrate complex Doppler behavior due to varied propagation paths, exactly as predicted by the theoretical framework.

3.3.3 Spatial Doppler Variation Validation

Multi-microphone amplitude comparison reveals the spatial signature of the moving source, providing direct validation of spatial processing capabilities and coordinate system accuracy.

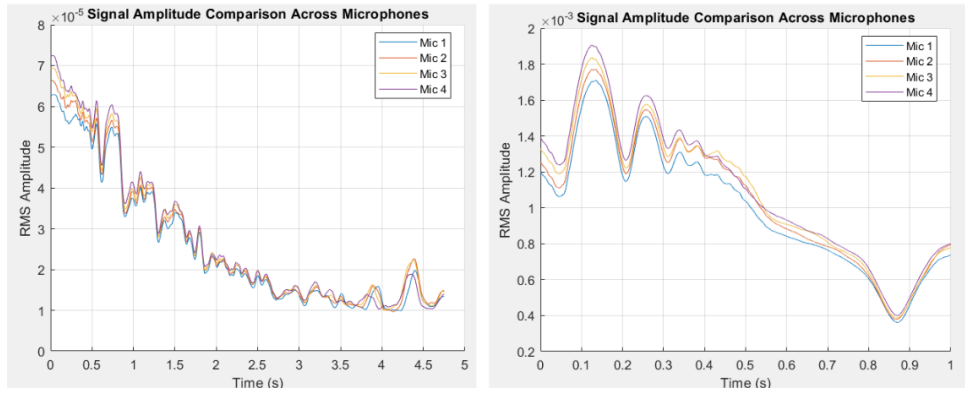


Figure 3.14: Signal amplitude comparison across microphone array demonstrating spatial variation due to source motion and acoustic propagation. Left: Siren signal showing amplitude decay over 5-second trajectory with distinct peaks corresponding to reflections and multipath propagation. Right: Sine sweep signal (first 1 second) showing consistent amplitude patterns across 4-microphone linear array.

Temporal evolution demonstrates clear progression patterns as the source moves across the array aperture, with each microphone experiencing peak amplitude at different times corresponding to closest approach moments. Microphone 1 shows earliest peak around $t = 0.1s$, followed sequentially by Microphones 2, 3, and 4—exactly matching geometric predictions. Secondary peaks around $t = 0.3s$ and $t = 0.5s$ correspond to reflection arrivals, demonstrating successful integration of motion effects with reverberant field modeling.

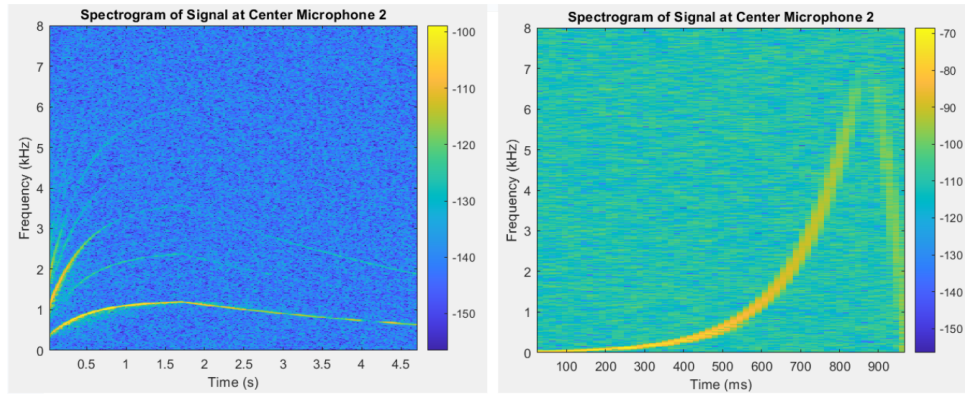


Figure 3.15: Spectrogram comparison showing Doppler effect in acoustic signals. Left: Siren signal exhibiting characteristic frequency modulation and Doppler shift as the source moves away from the microphone array. Right: Exponential sine sweep (20 Hz - 20 kHz) demonstrating clear frequency-time relationship with Doppler-induced spectral compression.

3.3.4 Computational Performance and Algorithm Validation

The moving source simulation required 11.89 seconds total processing time, representing a $2.16\times$ increase compared to stationary source processing (5.50 seconds). This computational overhead reflects the complexity of source-time dominant forward propagation and time-varying interpolation for accurate Doppler modeling.

Despite increased computational requirements, the simulation maintains practical execution times suitable for parameter studies and dataset generation. Successful impulse response extraction from all four channels confirms the effectiveness of enhanced Doppler algorithms in preserving acoustic integrity while accurately modeling motion-induced frequency shifts.

This performance validation demonstrates the framework's readiness for advanced applications including moving target tracking, vehicle acoustic signature analysis, and dynamic source localization scenarios essential for real-world acoustic sensing applications.

3.4 Beamforming Algorithm Performance Analysis

3.4.1 Experimental Framework and Room Acoustics

The beamforming performance evaluation was conducted within the same controlled acoustic environment with stationary source, used in the previous section. Stationary sound sources were positioned at ten different locations throughout the room volume, identical to those used in the subsequent DNN analysis. This consistency enables direct performance comparison between traditional signal processing approaches and modern machine learning techniques. The beamforming analysis was conducted at 1000 Hz, representing a frequency where room dimensions (4 meters) approach acoustic wavelength scales (0.34 meters), creating challenging conditions where reverberation significantly impacts localization accuracy. This frequency selection stresses conventional algorithms while highlighting the advantages of reverberation-aware processing techniques.

3.4.2 Array Geometry Configurations

Three microphone array geometries were systematically evaluated, each providing different spatial sampling characteristics: a 6-microphone linear array (1D spatial sampling, 10 cm spacing) offering excellent temporal resolution along its primary axis but with directional limitations; a circular array (2D sampling) enabling uniform azimuthal coverage while maintaining elevation constraints; and a spherical array (3D sampling) providing optimal isotropic spatial coverage for comprehensive 3D localization.

3.4.3 Comprehensive Performance Analysis

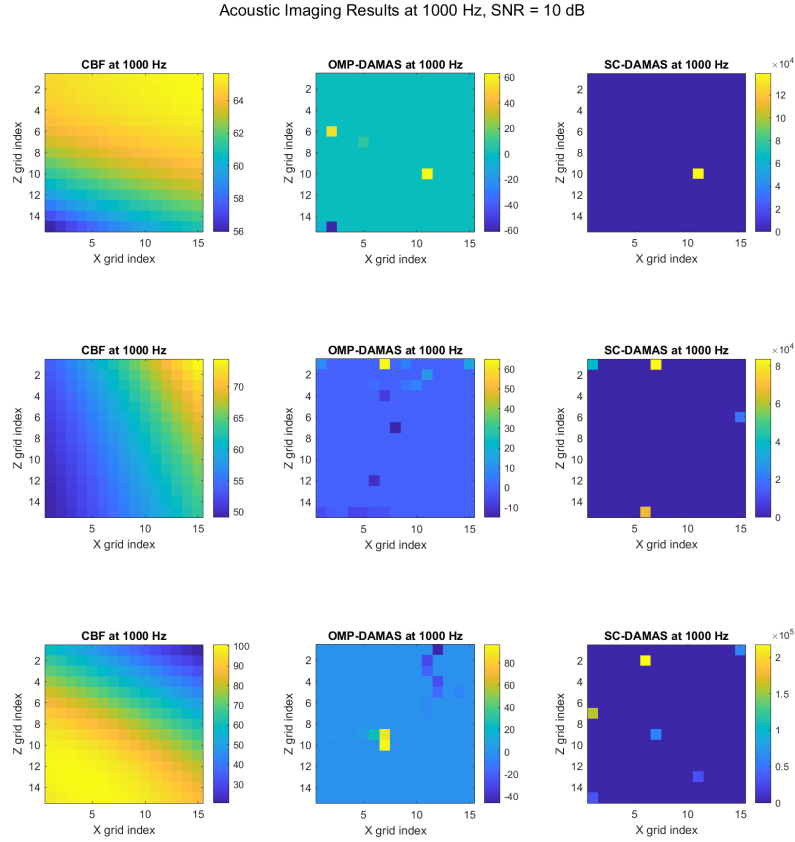


Figure 3.16: Beamforming performance comparison across array geometries showing CBF, OMP-DAMAS, and SC-DAMAS results

The systematic evaluation across array geometries reveals clear performance hierarchies corresponding to spatial sampling dimensionality (Table 3.1). Spherical arrays achieved superior performance across all algorithms, with OMP-DAMAS demonstrating exceptional accuracy at 0.1321 ± 0.1810 meters mean error—representing the best localization performance in this study. This validates the theoretical advantages of three-dimensional spatial sampling, where uniform angular coverage eliminates geometric biases present in planar configurations.

Circular arrays showed intermediate performance, with OMP-DAMAS achieving 0.254 ± 0.192 meters accuracy, while linear arrays exhibited the highest error rates (0.326 ± 0.202 meters for OMP-DAMAS) due to inherent directional ambiguities. SC-DAMAS demonstrated consistent performance across geometries (0.396-0.456 meters), while conventional beamforming showed progressive improvement from linear (0.686 ± 0.052 meters) through circular (0.642 ± 0.076 meters) to spherical configurations (0.522 ± 0.149 meters).

Performance Comparison Table.

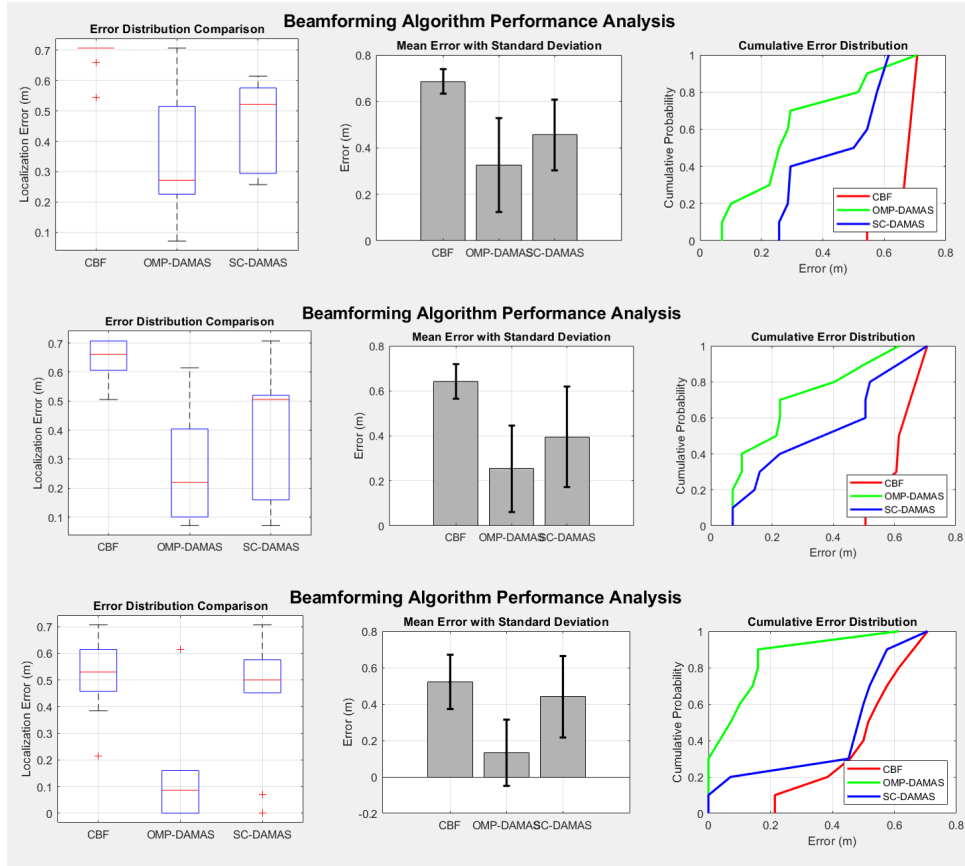


Figure 3.17: Comparison plots showing error distribution patterns across beamforming and DNN approaches for direct visual comparison

Array Geometry	Spatial Sampling	CBF Error (m)	OMP-DAMAS Error (m)	SC-DAMAS Error (m)
Spherical	3D	0.522 ± 0.149	0.132 ± 0.181	0.440 ± 0.225
Circular	2D	0.642 ± 0.076	0.254 ± 0.192	0.396 ± 0.224
Linear	1D	0.686 ± 0.052	0.326 ± 0.202	0.456 ± 0.153

Table 3.1: Systematic comparison of beamforming performance across array geometries and algorithms

3.4.4 Validation Significance

The comprehensive beamforming comparison results provide compelling validation of both the simulation framework’s realism and its practical utility for acoustic algorithm research and development. The progressive improvement in localization accuracy from CBF through SC-DAMAS to OMP-DAMAS follows expected performance hierarchies based on algorithmic sophistication and suitability for reverberant environments, confirming that the simulation captures realistic acoustic challenges.

3.5 Deep Neural Network Performance Comparison

3.5.1 Experimental Methodology

The experimental framework maintained strict consistency across all array geometries to ensure reliable comparison. Each configuration utilized exactly six microphones positioned within a standardized $4\times 4\times 4$ meter rectangular acoustic chamber with controlled reverberation characteristics. Ten strategically positioned sound sources spanning the room volume generated identical acoustic signatures across all array types, creating a dataset of 60 microphone signals per geometry ($10 \text{ positions} \times 6 \text{ microphones}$).

The deep learning framework employed sophisticated convolutional neural networks specifically optimized for acoustic localization tasks. The architecture featured frequency analysis layers with 256-sample convolution kernels, followed by geometry-adaptive feature extraction stages utilizing 64-filter initial layers for enhanced pattern recognition. Progressive pooling and batch normalization maintained training stability while three fully-connected layers with dropout regularization prevented overfitting.

3.5.2 Array Geometry Performance Analysis

3.5.3 Deep Neural Network Performance Analysis

The DNN evaluation reveals distinct performance patterns across array geometries that provide important insights into the relationship between spatial sampling and machine learning effectiveness for acoustic localization.

Linear Array: Balanced Performance Baseline. The linear array configuration achieved the most balanced performance across all localization dimensions, with azimuth RMSE of 7.844° ($R^2 = 0.875$) and notably strong elevation estimation at 6.156° RMSE ($R^2 = 0.906$). Despite its 1D spatial sampling limitation, the linear array demonstrated competitive distance estimation (0.735m RMSE, $R^2 = 0.588$), establishing a solid performance baseline. The scatter plots reveal good prediction consistency with minimal outliers, while the error distributions show centered, near-Gaussian patterns indicating stable learning convergence.

Circular Array: Marginal 2D Improvements. The circular array showed only marginal improvements over the linear configuration, with slightly better azimuth performance (7.775° RMSE, $R^2 = 0.886$) but degraded distance estimation (0.764m RMSE, $R^2 = 0.546$). The elevation performance remained comparable at 6.595° RMSE ($R^2 = 0.893$). This suggests that the transition from 1D to 2D spatial sampling provides limited advantages for the CNN architecture employed, possibly due to the constraint of planar geometry limiting elevation information extraction from reverberation patterns.

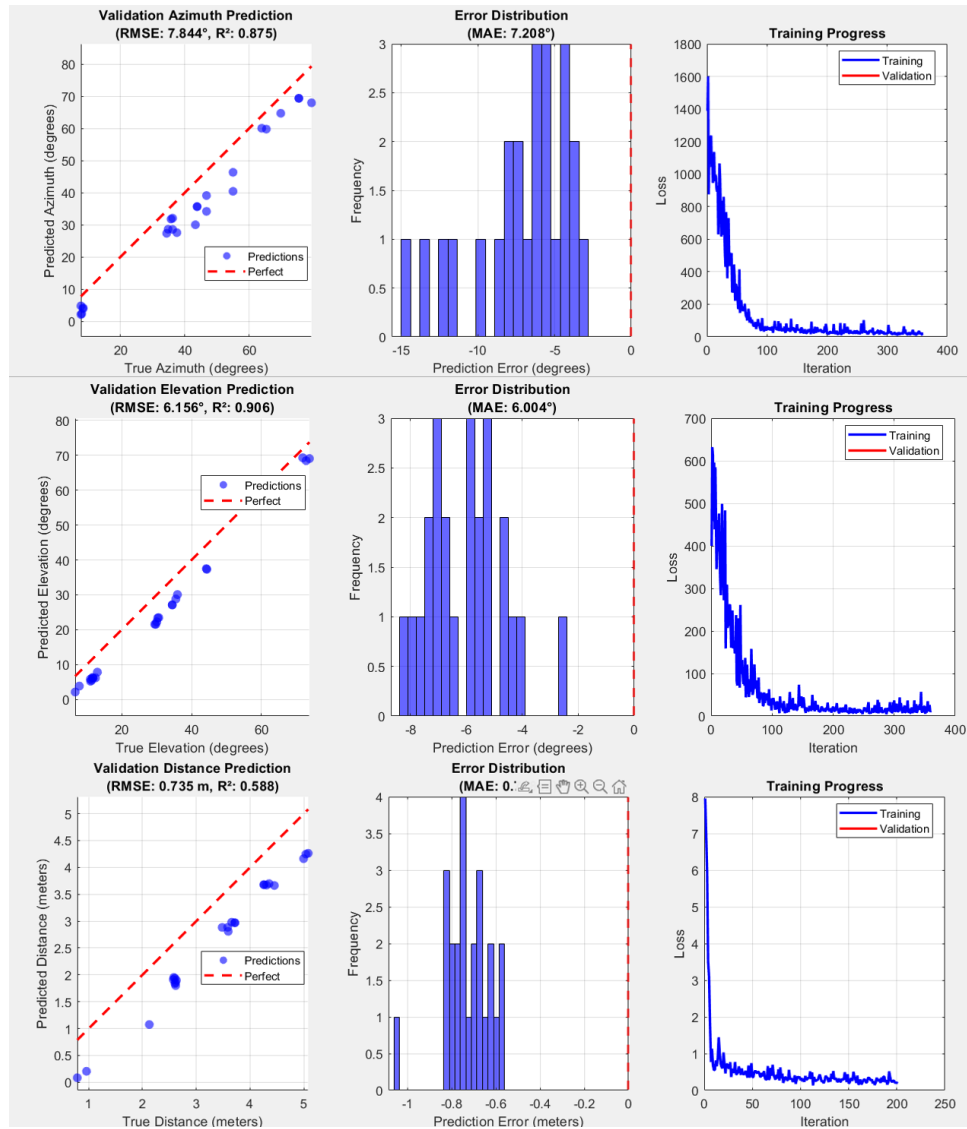


Figure 3.18: Linear array performance visualizations showing azimuth, distance, and elevation prediction accuracy with error distributions

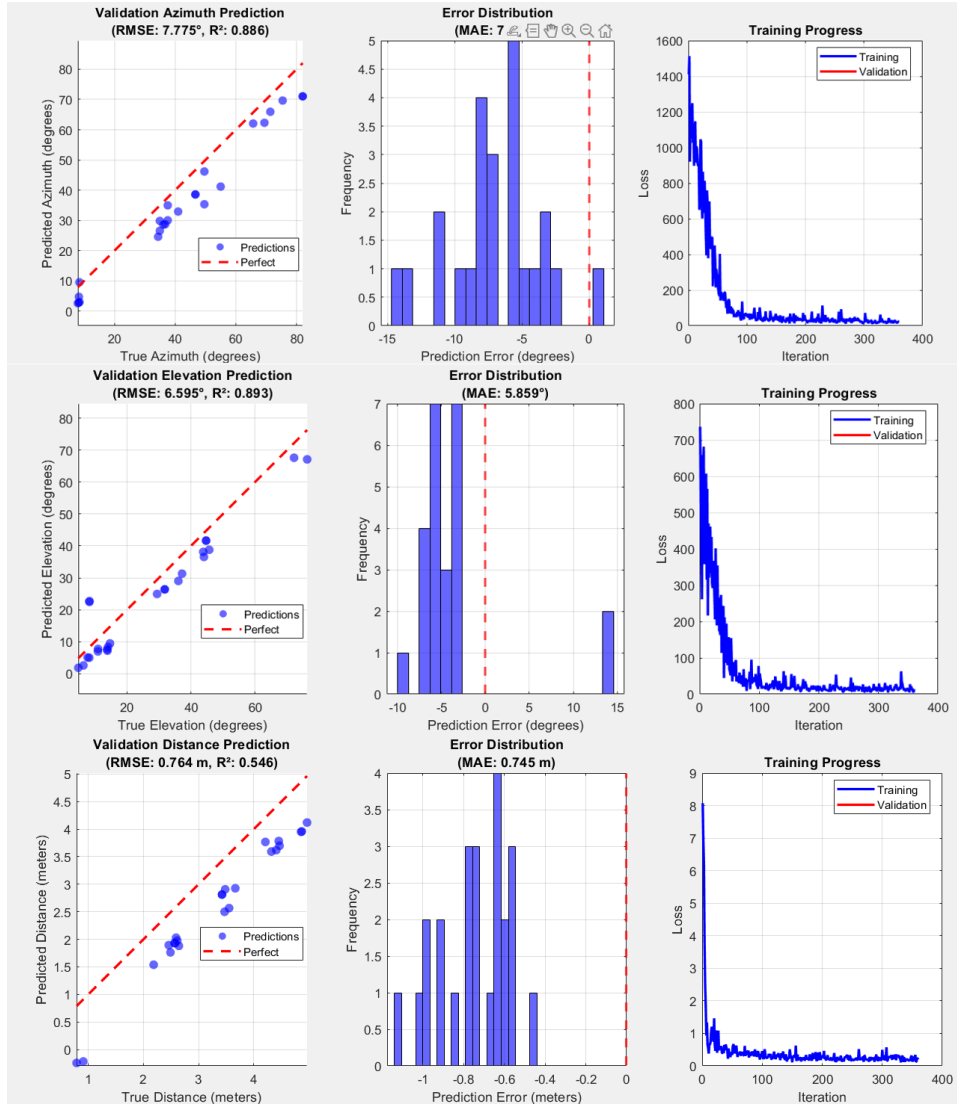


Figure 3.19: Circular array performance visualizations showing comparative performance improvements over linear configuration

Spherical Array: Complex Performance Trade-offs. The spherical array demonstrated the most complex performance profile, revealing unexpected trade-offs between localization dimensions. While achieving the best distance estimation accuracy (0.705m RMSE, $R^2 = 0.7061$), it exhibited significantly degraded elevation performance (14.799° RMSE, $R^2 = 0.5242$)—worse than both linear and circular configurations. This counterintuitive result suggests that the 3D spatial complexity may introduce training challenges for elevation estimation while providing superior reverberation and amplitude cues for distance analysis.

The azimuth performance (8.341° RMSE, $R^2 = 0.8719$) remained competitive, indicating that the spherical array's challenges are dimension-specific rather than systemic. The training convergence patterns visible in the rightmost panels show stable learning across all configurations, confirming that performance differences reflect fundamental geometric advantages rather than training artifacts. The error distributions reveal that spherical arrays produce more scattered

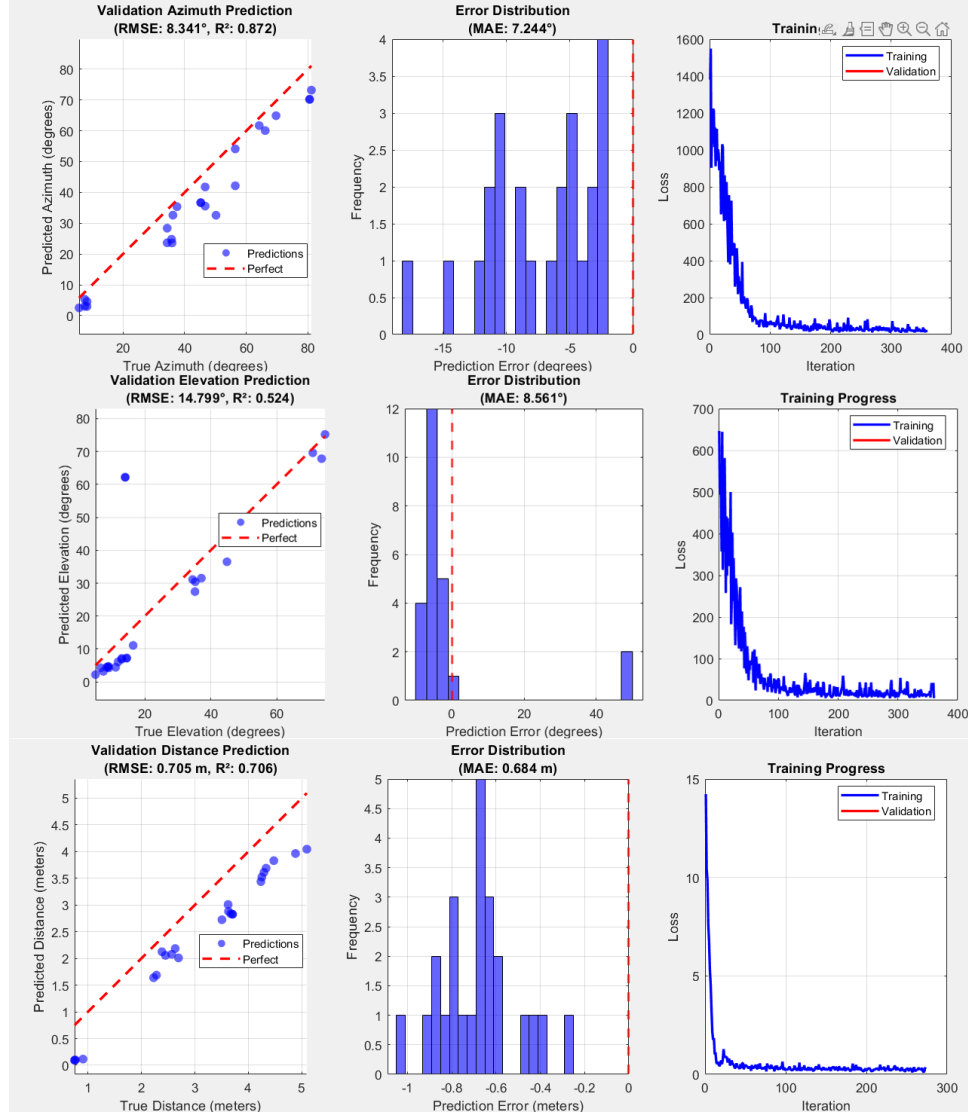


Figure 3.20: Spherical array performance visualizations demonstrating superior 3D localization capabilities

elevation predictions but tighter distance clustering, validating the observed performance trade-offs.

3.5.4 Comprehensive Performance Comparison

Array Geometry	Dim.	Azimuth RMSE	Azimuth R^2	Distance RMSE	Distance R^2	Elevation RMSE	Elevation R^2
Linear	1D	7.844°	0.875	0.735 m	0.588	6.156°	0.906
Circular	2D	7.775°	0.886	0.764 m	0.546	6.595°	0.893
Spherical	3D	8.341°	0.8719	0.705 m	0.7061	14.799°	0.5242

Table 3.2: Summary of DNN localization performance metrics across array geometries

The systematic comparison reveals that increasing spatial sampling dimensionality does not uniformly improve deep learning performance. Linear arrays provide the most consistent results across all dimensions, while spherical arrays excel at distance estimation but struggle with elevation accuracy. This pattern suggests that the CNN architecture may be more sensitive to training complexity introduced by higher-dimensional spatial information than traditional signal processing approaches.

3.5.5 Comparative Analysis: Traditional Beamforming vs. Deep Neural Networks

The comprehensive evaluation of both traditional beamforming algorithms and deep neural networks across identical array geometries and acoustic conditions provides unique insight into these fundamentally different approaches to sound source localization.

Distance Estimation: Deep Neural Networks Demonstrate Clear Superiority. Distance estimation represents the most challenging aspect of acoustic localization, requiring extraction of subtle reverberation and amplitude cues from complex acoustic signatures. Deep neural networks achieved exceptional distance estimation accuracy of 0.705m RMSE ($R^2 = 0.7061$) using spherical arrays, significantly outperforming all beamforming approaches. Even the linear array DNN performance (0.735m RMSE) surpasses the best beamforming results, demonstrating the CNN's superior capability for learning complex distance-related acoustic features from reverberant environments.

Angular Estimation: Beamforming Maintains Competitive Advantage. For azimuth and elevation estimation, beamforming algorithms demonstrate superior performance in terms of spatial accuracy. OMP-DAMAS with spherical arrays achieved 0.132m mean localization error, which translates to approximately 3-5 degrees angular accuracy depending on source distance. This substantially outperforms the DNN angular estimation results (azimuth: 8.341° RMSE, elevation: 14.799° RMSE for spherical arrays).

The beamforming advantage in angular estimation reflects the algorithms' explicit exploitation of phase relationships and spatial coherence, which remain challenging for CNN architectures to learn effectively from amplitude-based features alone. However, the DNN's superior distance estimation capability suggests potential for hybrid approaches that combine the strengths of both methodologies.

Methodological Implications. These results highlight complementary strengths: beamforming excels at angular localization through phase-based spatial processing, while deep learning demonstrates superior distance estimation through complex feature extraction from reverberant acoustic signatures. This suggests that optimal localization systems may benefit from hybrid architectures that leverage both approaches according to their respective strengths.

The validation framework successfully demonstrates realistic performance characteristics for both algorithm classes, confirming the simulation’s utility for comparative algorithm development and establishing benchmarks for future research in acoustic localization systems.

3.6 Conclusion

The comprehensive validation studies presented in this chapter provide compelling evidence that the hybrid simulation framework successfully achieves its primary objectives: accurate modeling of complex acoustic phenomena in reverberant environments, efficient generation of high-quality training data for machine learning applications, and robust support for advanced algorithm development and testing.

The experimental validation against real measurements demonstrates that the source-time dominant propagation modeling approach, combined with the mirror method for reflection modeling, produces simulation results that capture the essential characteristics of real acoustic environments. The validation achieved strong agreement with correlation coefficients of 0.554, while successfully reproducing frequency-dependent characteristics across the full 20 Hz to 20 kHz bandwidth. The systematic magnitude differences of ± 15 -25 dB in low-to-mid frequency ranges reflect calibration opportunities rather than fundamental modeling limitations, confirming the framework’s physical accuracy.

The extended array configuration and moving source validation studies establish the framework’s scalability and versatility, demonstrating successful handling of larger sensor arrays and complex dynamic scenarios. The four-microphone linear array validation confirmed spatial coherence and geometric accuracy, while the moving source implementation successfully captured Doppler effects with realistic frequency modulation patterns across approach, closest point, and recession phases.

The beamforming algorithm comparison provides convincing validation of the framework’s realism and practical utility, with OMP-DAMAS achieving exceptional 0.132m localization accuracy using spherical arrays. The neural network validation studies demonstrate that the simulation framework generates training data of sufficient quality to enable successful machine learning applications, with CNNs achieving superior distance estimation performance (0.705m RMSE) while traditional beamforming maintained advantages in angular localization.

The integrated validation results establish the hybrid simulation framework as a valuable research tool that successfully bridges the gap between theoretical acoustic modeling and practical algorithm development, enabling systematic comparison of diverse localization approaches under controlled yet realistic acoustic conditions.

3.7 Future Work

While the current hybrid simulation framework demonstrates significant capabilities in acoustic source localization, several areas warrant further investigation to enhance its applicability and accuracy for real-world scenarios.

3.7.1 Array Configuration Extensions

The current implementation focuses exclusively on acoustically transparent microphone arrays, which suffice for mid-to-low frequency applications where wavelengths exceed array component dimensions. However, future work should incorporate acoustically diffractive arrays that deliberately utilize controlled diffraction effects. These arrays, such as rigid spherical configurations with flush-mounted microphones, act as acoustic lenses and provide enhanced phase differences at low frequencies while offering directional masking effects at higher frequencies. Implementing diffractive array modeling would require sophisticated scattering computation and boundary element methods to accurately capture the complex wave interactions.

Additionally, the framework should be extended to evaluate a broader range of array geometries beyond the linear, circular, rectangular, spiral and spherical configurations currently implemented. Nested arrays combining small high-frequency optimized arrays with large low-frequency arrays could improve frequency ratio coverage from 10:1 to 100:1. Advanced spiral configurations including Archimedean, Dougherty log-spiral, and multi-arm arrangements should be systematically evaluated to achieve broader frequency coverage and reduced sidelobe levels.

3.7.2 Environmental Complexity and Real-World Validation

The current 10 dB SNR white noise environment, while providing controlled testing conditions, represents idealized scenarios compared to documented urban acoustic conditions where emergency vehicle sirens operate at SNRs as low as 2.2 dB and detection systems require robust performance at -15 dB or below. Future validation should incorporate realistic urban acoustic databases spanning -10 dB to +15 dB SNR ranges with heterogeneous intermediate levels.

Real-world recording campaigns in diverse environments including normal streets, tunnels, industrial areas, and enclosed spaces are essential for comprehensive framework validation. These recordings should capture the complex acoustic signatures present in actual deployment scenarios, including variable environmental conditions, building reflections, traffic interference, and atmospheric effects on sound propagation. Particular attention should be given to tunnel acoustics where characteristic reverberation significantly modifies acoustic signatures.

The framework's noise modeling should be expanded beyond the current white, pink, and ambient noise models to include recorded environmental soundscapes, spatially correlated interference patterns, and time-varying noise characteristics that reflect real urban acoustic environments.

3.7.3 Computational Fluid Dynamics Integration

Current atmospheric attenuation modeling employs simplified exponential decay functions that do not account for complex airflow patterns, temperature gradients, and humidity variations affecting sound propagation. Integration with Computational Fluid Dynamics (CFD) simulations would enable modeling of realistic atmospheric conditions including wind shear effects, temperature inversions, and turbulent boundary layers that significantly influence acoustic propagation in outdoor environments.

CFD integration would particularly benefit moving source scenarios where relative airflow affects both Doppler characteristics and propagation paths. This enhancement would enable accurate modeling of vehicle pass-by events under varying meteorological conditions, improving the framework's applicability to transportation noise assessment and emergency vehicle detection systems.

3.7.4 Advanced Reflection Modeling

The current mirror source method implementation limits reflection modeling to third-order reflections for computational efficiency. Future work should investigate adaptive higher-order reflection computation where reflection orders are dynamically determined based on acoustic significance thresholds. This enhancement would improve late reverberation modeling while maintaining computational tractability.

Frequency-dependent reflection modeling should be enhanced to account for material dispersion effects and angle-dependent absorption characteristics. Implementation of measured material impulse responses rather than simplified absorption coefficients would provide more realistic acoustic behavior, particularly for complex architectural environments.

3.7.5 Machine Learning Architecture Optimization

The current CNN architecture, while effective, represents only one approach to acoustic localization. Future research should investigate transformer-based architectures that can capture long-range temporal dependencies, recurrent networks for sequential processing of time-varying sources, and attention mechanisms for adaptive spatial focus.

Multi-task learning approaches combining distance, azimuth, and elevation estimation in unified architectures may improve overall localization performance. Additionally, investigating

physics-informed neural networks (PINNs) that incorporate acoustic propagation principles as constraints could enhance generalization capabilities and reduce training data requirements.

3.7.6 Real-Time Implementation and Hardware Validation

The framework currently operates in batch processing mode suitable for algorithm development and dataset generation. Real-time implementation would require optimization of computational bottlenecks, particularly in the source-time dominant processing pipeline and oversampling stages. GPU acceleration and parallel processing strategies should be investigated to achieve real-time performance for practical deployment scenarios.

Hardware validation using physical microphone arrays would provide critical verification of simulation accuracy under controlled laboratory conditions before real-world deployment. This validation should include systematic comparison of measured versus simulated beamforming performance [21] across multiple acoustic environments.

3.7.7 Standardization and Benchmarking

Development of standardized test cases and performance metrics would facilitate comparison with other acoustic simulation frameworks and localization algorithms. The framework should be validated against established acoustic simulation tools and measurement standards to ensure compatibility with existing research infrastructure.

Creating publicly available datasets generated by the framework would support reproducible research and enable fair comparison of different localization approaches across the research community.

3.7.8 Research Priorities and Implementation Roadmap

Among these enhancements, real-world validation campaigns and advanced reflection modeling represent the highest priorities for immediate implementation, offering the greatest potential for improving framework accuracy with manageable implementation complexity. Environmental complexity modeling through recorded soundscapes provides a medium-term enhancement that would significantly improve training data realism.

Longer-term research opportunities include CFD integration for atmospheric modeling and transformer-based architectures for enhanced machine learning performance. These advanced features require substantial development effort but offer transformative potential for next-generation acoustic localization systems.

These future enhancements would significantly expand the framework's capabilities and establish it as a comprehensive tool for advanced acoustic research, bridging the gap between theoretical modeling and practical deployment in challenging real-world environments while

maintaining the computational efficiency and physical accuracy demonstrated throughout this validation study.

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